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San3a Application (Handcrafter App)

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Academic Year 2024/2025
Final Year Document

Abstract

This project presents a comprehensive handmade application designed to bridge the gap between artisans and customers, offering a unique platform for custom handmade products, AI-driven recommendations, and interactive features. The application addresses the growing demand for personalized, eco-friendly, and unique handmade goods by providing a suite of innovative tools and features. The primary purpose of this project is to create a user-friendly platform that enhances the shopping experience for customers while empowering artisans to showcase their craftsmanship and connect with potential buyers. The significance of this project lies in its ability to revolutionize the handmade goods market by leveraging AI and user-centric design to create a seamless, interactive, and eco-conscious shopping experience. The results of this research are useful for both consumers seeking personalized, high-quality products and artisans looking to expand their reach and streamline their business operations. By addressing the challenges of modern e-commerce, this application fosters a vibrant community of creators and buyers, promoting sustainability, creativity, and innovation in the handmade industry.

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List of Abbreviations

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Chapter 1: Introduction

1.1 Background and Motivation

In recent years, there has been a significant shift in consumer preferences toward personalized, unique, and eco-friendly products. Handmade goods, in particular, have gained immense popularity due to their authenticity, craftsmanship, and sustainability. However, artisans and small-scale creators often face challenges in reaching a broader audience, managing custom orders, and competing with mass-produced items. On the other hand, customers seeking handmade products struggle to find reliable platforms that offer a wide variety of options, personalized recommendations, and seamless communication with artisans.

The rise of e-commerce and advancements in artificial intelligence (AI) have created opportunities to bridge this gap. While platforms like Etsy and Amazon Handmade have made strides in connecting artisans with buyers, there is still a need for a more comprehensive, user-friendly, and feature-rich application tailored specifically to the handmade industry. This project aims to address these challenges by developing a dedicated platform that not only facilitates the buying and selling of handmade products but also enhances the overall experience for both artisans and customers.

Motivation:

1. **Empowering Artisans:** Many talented artisans lack the resources or technical expertise to market their products effectively. This application provides them with a platform to showcase their work, connect with customers, and manage custom orders efficiently. Features like Notifications for Sellers and Add Offers on the Post enable artisans to respond quickly to customer requests, increasing their chances of securing business.
2. **Personalized Shopping Experience:** Customers today seek unique, customized products that reflect their individuality. The AI-Driven Gifting Feature, Style Advice AI, and AI To Design tools cater to this demand by offering personalized recommendations and design ideas. The Search by Photo feature further simplifies the process of finding products that match the user's preferences.
3. **Sustainability and Eco-consciousness:** With growing awareness of environmental issues, consumers are increasingly drawn to eco-friendly products. The Eco-friendly Flag highlights sustainable items, helping users make informed choices that align with their values.
4. **Convenience and Efficiency:** The application streamlines the shopping process with features like Express Delivery Flag, Comparison Tool, and Recommendation

System, saving users time and effort. The Event Reminder System and Wishlist add convenience, while the Points system incentivizes repeat purchases.

5. **Community Building:** The platform fosters a sense of community by enabling direct communication between artisans and customers through the Chat feature. The Online Trade Show and Bazar on Application create opportunities for artisans to showcase their work and engage with a wider audience.
6. **Supporting Local Economies:** By promoting handmade products and hosting Ads for Offline Bazar, the application supports local artisans and small businesses, contributing to the growth of local economies.
7. **Innovation in E-commerce:** This project leverages cutting-edge AI technologies to create a unique and interactive shopping experience. Features like AI-driven gifting, Style Advice AI, and AI to Design demonstrate the potential of AI to transform the handmade goods market.

1.2 Problem Statement

- In today's market, artisans often struggle with limited exposure and the exploitation of their work by shop owners, who take a significant portion of their profits. Additionally, customers seeking custom handcrafted products often face challenges in finding artisans who can accurately translate their ideas into tangible items.
- The current marketplace lacks an efficient, user-friendly platform that connects artisans directly with customers and empowers both parties. Artisans need a solution to showcase their talents, receive fair compensation, and gain exposure, while customers need a seamless way to communicate their product desires and receive personalized handcrafted items.

1.3 Objectives

- Our e-commerce helps to connect artisans and Clients by providing a platform for artisans to showcase and sell handcrafted products directly to clients.
- we help customers make product requests by allowing clients to find or create products by uploading images or describing their vision for artisans to fulfill.
- we try to help the customer with personalized gifts by offering tailored gift recommendations based on client-provided information about the recipient.
- we are looking forward to increasing Artisan Income, enabling artisans, including stay-at-home individuals, to generate additional income through their crafts.
- we care about boosting artisan visibility by promoting artisans with tools like media content, reviews, and workshop ads to build trust and attract clients.
- because we are looking for customer and artisan comfort, we provide flexible payments and offers, we provide artisans the tools to create special offers and ensure clients have flexible payment options.

1.4 Project Scope and Limitations

- **Scope:**
 - Development of a mobile application with 26 features, including AI-driven tools, ethical shopping flags, and interactive social features.
 - Target audience: Artisans and customers in the handmade goods market.
 - Focus on personalization, customization, and ethical shopping.

- **Limitations:**

- Technical complexity of AI features like photo search and gifting recommendations.
- Resource constraints, including budget and development expertise.
- Competition from established e-commerce platforms.
- Dependence on third-party logistics for delivery

- **Project Plan**

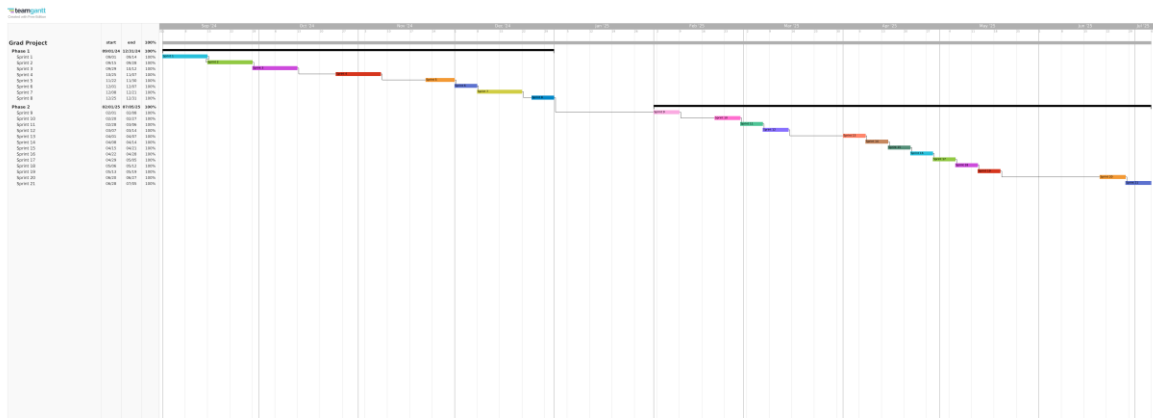


Figure 1. Project Plan

- **Team Responsibility**

Name	Responsibility
Maryam Abdelhamed	Frontend (Flutter)
Alaa Hossam	Frontend (Flutter)
Asmaa Saleh	UI/UX, Testing
Ammar Mahmoud	Backend (Nest)
Mina Emad	Backend (Nest)
Abdelrahman Ashraf	AI Developer

Table 1. Team Responsibility

1.5 Project Methodology

- **Mixed Approach (Waterfall and Agile)** We used a mixed approach combining Waterfall and Agile methodologies as our project life cycle. This hybrid approach leverages the structured planning of Waterfall and the flexibility of Agile to ensure a balanced and efficient development process.
 - ✓ **Initial Planning and Requirements Gathering (Waterfall):** The project will begin with a detailed requirements analysis, including functional requirements, user personas, and use cases. This phase will also include a SWOT analysis, gap analysis, and feasibility study to ensure the project is well-defined and achievable. A comprehensive project plan will be created, outlining the timeline, milestones, and deliverables.
 - ✓ **Iterative Development (Agile):** Once the initial planning is complete, the development process will shift to Agile. The project will be broken into smaller sprints, each focusing on delivering specific features (e.g., AI-driven gifting, photo search, eco-friendly flags). At the end of each sprint, the team will review progress, gather feedback, and make necessary adjustments. This iterative process ensures that the product evolves based on real user needs and feedback.
 - ✓ **Testing and Deployment (Agile):** deployment and testing will follow an Agile approach, utilizing Continuous Integration and Continuous Deployment (CI/CD) practices. This allows for rapid iterations and quick feedback, ensuring that the product is consistently updated and improved post-launch.

Chapter 2: Market and Literature Survey

2.1 Competitors

Feature	San3a	Amazon	Alibaba	Etsy	Handmade.com
AI-driven Gifts	✓	✗	✗	✗	✗
Customized product	✓	✗	✗	✗	✗
Seller Customized Profile	✓	✗	✗	✗	✗
Search by Photo	✓	✗	✓	✗	✗
Chat	✓	✗	✓	✓	✗
Comparison Tool	✓	✗	✗	✗	✗
Handicrafter analysis	✓	✓	✓	✓	✓
Online Bazar Section	✓	✗	✗	✗	✗
Advertising	✓	✓	✓	✓	✓
Reels for Sellers	✓	✗	✗	✗	✗
Warehouse For Handmade products	✗	✗	✓	✗	✗

Table 2: competitor table

San3a stands out in the competitive landscape for several key reasons:

1. Comprehensive Feature Set:

- **Search by Photo:** Unlike many competitors, San3a allows users to search for products using images, making finding exactly what they want easier.
- **AI-driven Gifts:** Our AI technology enhances user experience by recommending personalized gifts, a feature not offered by most competitors.
- **Event Reminder:** This unique feature helps users remember important occasions, ensuring they never miss an opportunity to celebrate.

2. Enhanced Wishlist Functionality:

- Users can create and share wishlists, which San3a and some competitors support. However, our user-friendly interface and additional features like gift recommendations make the experience more engaging.

3. Robust Communication Tools:

- **Chat Feature:** San3a offers an integrated chat function, allowing users to communicate directly with sellers, which enhances trust and engagement. Other platforms lack this direct line of communication.
- 4. **Gift Card Options:**
 - The availability of gift cards provides flexibility for users, allowing them to give the gift of choice, which is not universally available among competitors.
- 5. **Advanced Comparison and Recommendation Tools:**
 - **Comparison Tool:** Users can compare products easily, helping them make informed decisions. This feature is absent in many competing platforms.
 - **Recommendation System:** Our system not only recommends products but can also suggest the cheapest options, saving users money.
- 6. **User Engagement Features:**
 - **Points System:** Users earn points for purchases, encouraging loyalty and repeat business. This gamification aspect is not prevalent among competitors.
 - **History of Watched Products:** San3a keeps track of user interests, making it easier for them to revisit and purchase items they like.
- 7. **Eco-Friendly Initiatives:**
 - The Eco-Friendly Flag showcases our commitment to sustainability, appealing to environmentally-conscious consumers, a feature not highlighted by competitors.
- 8. **Customization and Personalization:**
 - **Seller Customizable Profile:** This feature allows sellers to create a more personal connection with buyers, fostering a community feel that is often missing in larger platforms.
- 9. **Support for Local Businesses:**
 - **Online Bazaar Section:** San3a supports local artisans and small businesses, promoting community engagement and unique products that are not available on larger, impersonal platforms.
- 10. **Flexible Delivery Options:**
 - **Special Order (Faster Delivery) and Express Delivery Flag:** These options cater to users' need for speed and reliability in delivery, enhancing overall customer satisfaction.

2.2 Surveys

2.2.1 Online Surveys

- **Aytm.com:**

- They asked 1,000 respondents on August 9-11-2016 about their thoughts on the market for handmade products. There's a rising demand for handmade items in the gift industry.

Of those who have purchased handmade items, 53% have done so at craft fairs. 39% have bought handmade items at flea markets. 30% have bought them from local stores. 27% have bought them from farmer's markets. 25% have purchased handmade products on eBay and another 25% on Etsy. 20% have purchased handmade items on Amazon. 10% have bought from independent online stores. 6% have bought from other marketplace sites. And 5% have purchased handmade items from other sources, like from friends directly or through Facebook.



Figure 2: survey about handmade shopping source

In addition, 51% of handmade shoppers said that they have purchased handmade fashion accessories. 46% have purchased home items. 42% have bought holiday items or decorations. 39% have bought artwork or prints. 34% have bought food items. 30% have bought handmade clothing. 17% have bought toys or kids' items from handmade sellers. And 5% have purchased other types of products, such as cards or paper goods.



Figure 3: Handmade shopping items

In general, 48% of respondents said handmade shopping has become more popular in the past five years. 49% think that handmade shopping is likely to continue gaining popularity over the next five years. And 47% said they are at least somewhat likely to purchase handmade items over the next year.



Figure 4: Handmade popularity

Of those who have purchased handmade items, 81% said they chose to buy handmade because they could get truly unique items. 51% said they like buying handmade items to support small businesses. And 30% cited cost savings when buying handmade items. Overall, 72% of all respondents think handmade items are more special than mass-produced items. 58% think that handmade items are of higher quality than mass-produced items. And 55% think that handmade items are more stylish than mass-produced items.

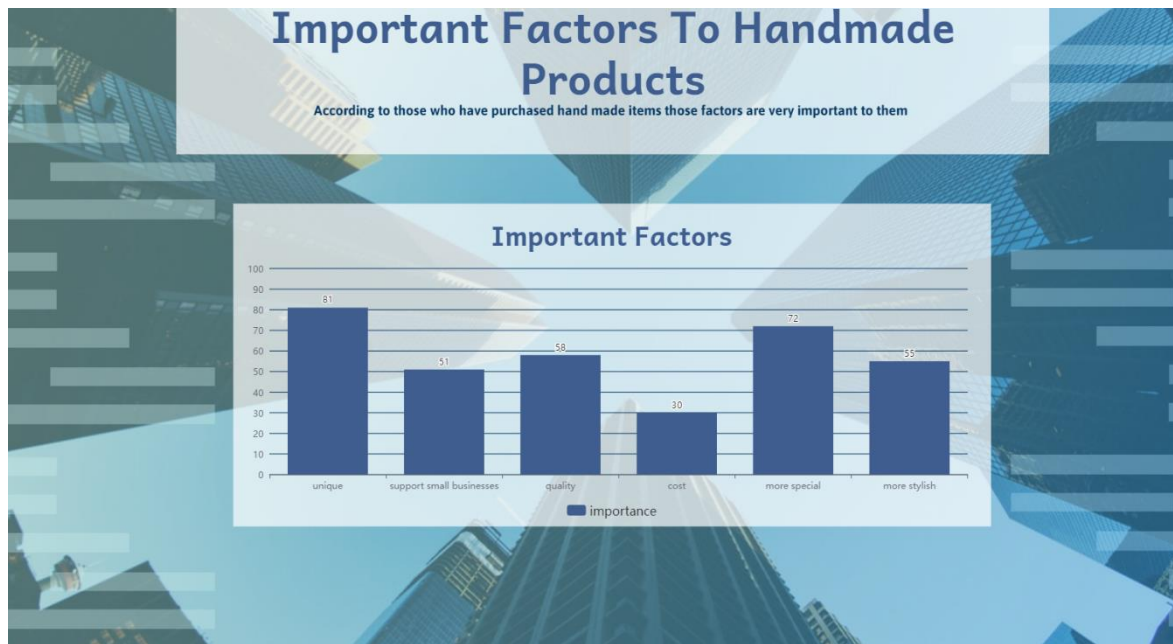


Figure 5: Important factors in handmade products

2.2.2 Our Survey

- We made a survey To Know what the handcrafters and customers really need

DO YOU LIKE HANDMADE PRODUCTS?

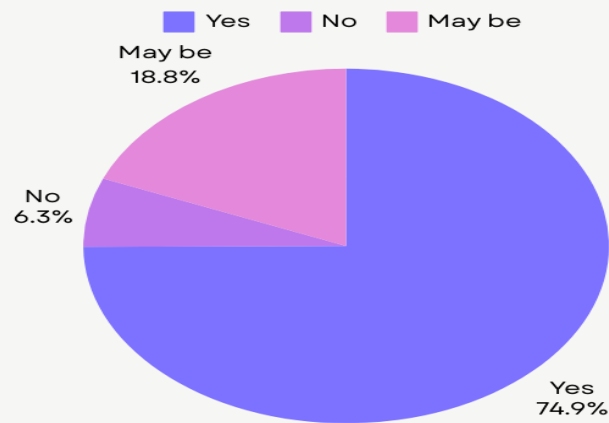


Figure 6: Survey Q1

IN YOUR OPINION, WHY DO MOST HANDICRAFT OWNERS NOT TAKE THEIR WORK IN BUSINESS WAY?

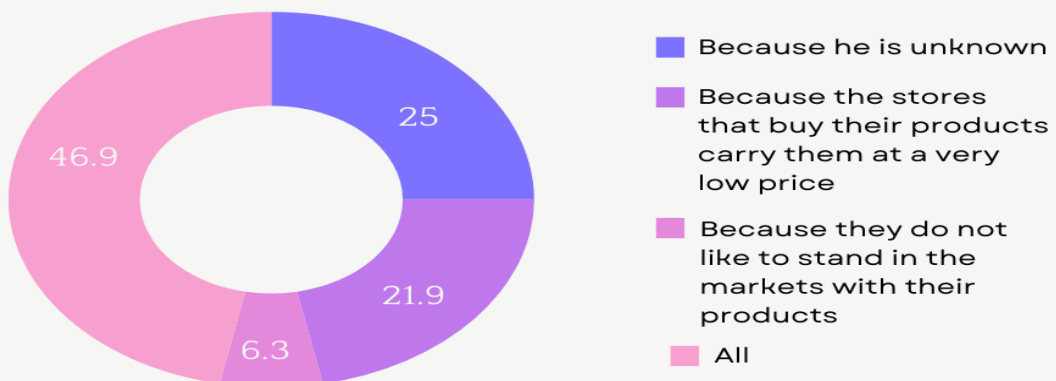


Figure 7: Survey Q2

HAVE YOU EVER WANTED TO SHARE YOUR WISH LIST FROM A WEBSITE WITH A FRIEND AND YOU COULDN'T?

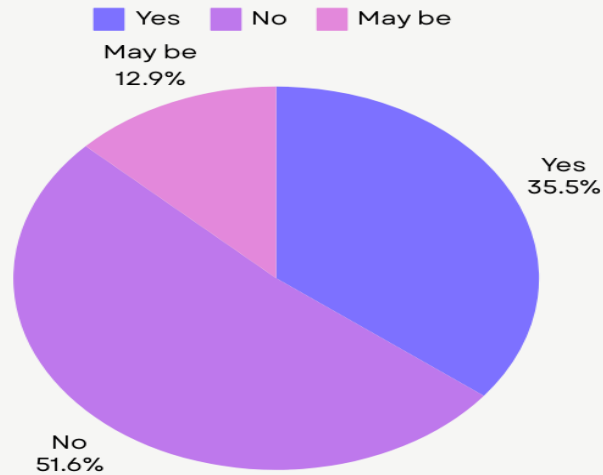


Figure 8: Survey Q3

DO YOU CARE IF THE PRODUCTS ARE HARMFUL TO THE ENVIRONMENT OR NOT?

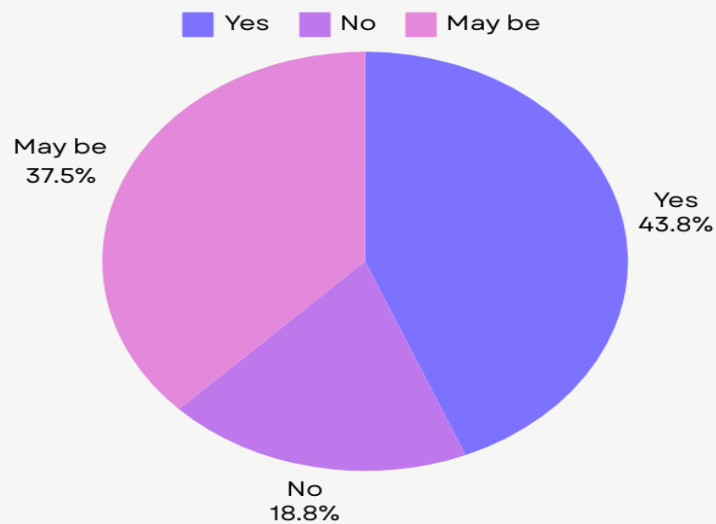


Figure 9: Survey Q4

WHEN YOU WANT TO MAKE A SPECIFIC DESIGN AND YOU SEARCHED A LOT IN PLACES THAT SELL THIS PRODUCT AND DID NOT FIND THE SAME DESIGN. IF THERE IS A TOOL THAT CAN PUT THE DESIGN IN YOUR IMAGINATION IN THE FORM OF AN IMAGE. IN YOUR OPINION, DOES THIS SAVE EFFORT, TIME AND MONEY?

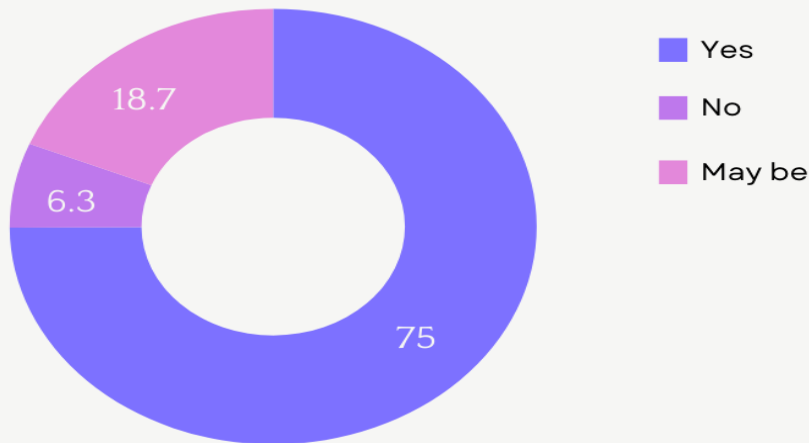


Figure 10: Survey Q5

IF BAZAARS BECOME AVAILABLE ONLINE AND CRAFTERS DISPLAY THEIR PRODUCTS AND YOU CHOOSE WHAT YOU LIKE FROM YOUR HOME, IS THAT BETTER FOR YOU?

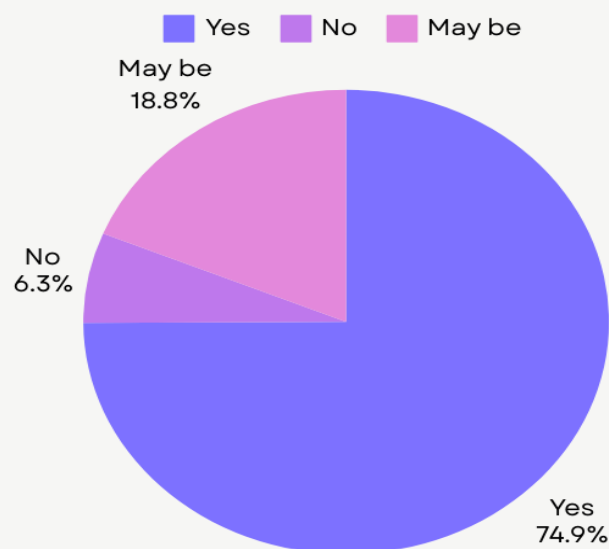


Figure 11: Survey Q6

WHEN YOU WANT TO BUY A GIFT FOR SOMEONE, DO YOU NEED SOMEONE'S HELP IN CHOOSING THE GIFT?

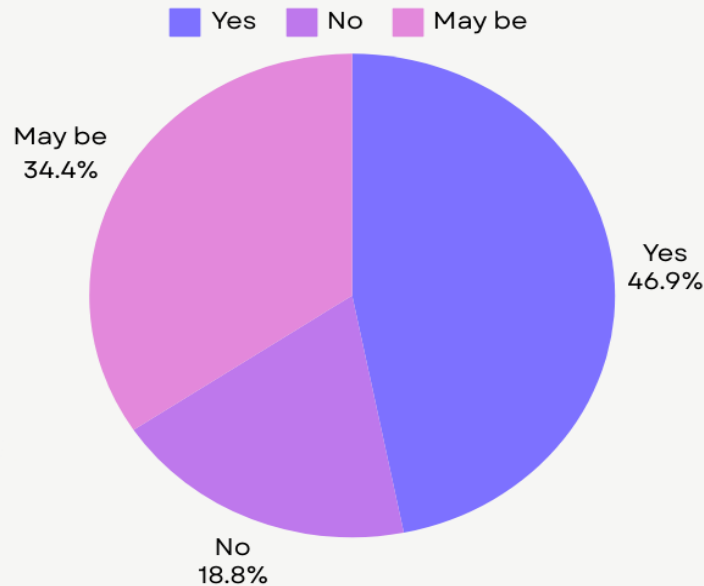


Figure 12: Survey Q7

Conclusion:

Based on the insights from the above regarding the survey on handmade products, here's an analysis of what features should be prioritized in the app and the rationale for each:

Key Features to Implement First

1. Search and Filter Options
2. Secure Payment Options
3. Reviews and Ratings
4. Customization Options
5. Sales Analytics for Handcrafters
6. Advanced Comparison and Recommendation Tools
7. AI-driven Gifts
8. Online Bazaar Section
9. Eco-Friendly Initiatives

2.3 SWOT Analysis

➤ Strengths:

- ✓ Variety of unique Services: Search by Photo, AI-driven gifting, and features specifically for handmade products like style advice AI.
- ✓ Ethical Shopping: Eco-friendly and boycott flags support users' ethical purchasing choices.
- ✓ Engagement Tools: Event reminders and points keep customers engaged.
- ✓ Advanced Search and Comparison Tools: Search by photo and product comparison tools to help customers make informed decisions enhance the user experience.
- ✓ Personalization: AI and recommendation tools enhance user experience by making shopping more personalized.
- ✓ Unique Seller Customization: Sellers can customize their profile, providing them with a unique way to show their brand.
- ✓ Interactive Social Features: chat, gift cards, reels, and wish lists sharing enhances social commerce capabilities.
- ✓ Encourage customer to buy: give customers points every time there buy anything which they can exchange this buy other products.
- ✓ Enhancing customer confidence: give customers general feedback for the seller they contact with which make them comfort.

➤ Weakness:

- ✓ Complexity: Managing a wide range of features such as AI tools, personalized gifting, customizable profiles could make development more complex.
- ✓ User Awareness: Educating users about innovative features (e.g., AI-driven gifting) could be difficult without proper guidance and support.

➤ **Opportunities:**

- ✓ Demand for Ethical Shopping: Eco-conscious consumers are seeking platforms that align with their values, creating a strong opportunity for our eco-friendly feature.
- ✓ Customization and Personalization: Consumers are seeking unique, personalized products, and services, also sellers can customize their own profiles which our platform can provide.
- ✓ Increased Social Commerce: Features like chat, reels, sharing wish lists and send a gift cards tap into the growing trend of social shopping.
- ✓ AI Integration: Development of new AI-driven features, including recommendations and visual search and style advice AI creating a strong opportunity for our platform.
- ✓ Cost Advantage: Potential to attract sellers with lower fees.
- ✓ Cheaper product recommendations: Leveraging recommendations for cheaper products to attract budget conscious shoppers.
- ✓ Event and Bazar Advertising: Promoting offline and online bazars will create new marketing opportunities for sellers.

➤ **Threats:**

- ✓ Competition: Intense competition from established e-commerce giants.
- ✓ Technology Risks: Relying on AI and advanced features can introduce potential technical challenges that affect the user experience.

2.4 Gap Analysis

➤ Current State:

- ✓ Limited features for handmade and bazar sections both online and offline are uncommon in e-commerce platforms.
- ✓ AI integration is minimal or limited to basic product recommendations.
- ✓ Ethical shopping features eco-friendly does not commonly exist.
- ✓ Platforms rarely provide interactive social features such as wish list sharing, reels, and chat. ✓ Lack of personalization and customization tools, especially in areas like handmade products, AI-driven gifting, and AI to design.
- ✓ Sellers have little control over customizing their online store page.
- ✓ Few platforms offer photo search functionality, making product discovery less intuitive for users who prefer visual search methods.
- ✓ Limited options for personalized notifications for sellers or users based on their interests.

➤ Desired future State:

- ✓ **Enhance Handmade Product Offerings & bazaars:**
 - Create a specialized handmade section.
 - Introduce an online bazaar and offline bazaar ads to provide unique selling opportunities.
- ✓ **Ethical Shopping Features:**
 - Introduce eco-friendly to help users make ethical purchasing decisions.
- ✓ **AI-driven personalization and customization:**
 - AI for personalized recommendations, gifting suggestions, and style advice to create a unique and tailored shopping experience and AI for customizing profiles.
- ✓ **Advanced Search and Comparison Tools:**
 - Implement advanced search functionality like photo search and comparison tools to make product discovery more efficient.
- ✓ **Interactive and Social Shopping:**
 - Add social features such as sharing wishlists and gift cards, Encouraging engagement through event reminders, a trend section, and a bazaar section on the app.
- ✓ **AI-Powered Features:**
 - Introduce an AI-driven gifting and recommendation system.

- ✓ **Increased Engagement:**
 - Use tools such as notifications for sellers, interest-based recommendations, event reminder, and points systems to improve user retention and activity
- ✓ **Faster and Special Delivery Options:**
 - Offer express delivery and a special order (faster delivery) option to cater to users looking for faster shipping.

➤ **GAP**

1. Bazar and Offline Ads:

- ✓ Gap: Lack of platforms offering an online bazar or offline bazar ads.
- ✓ Solution: Develop online bazar sections and offline advertising options for sellers.

2. Ethical Shopping:

- ✓ Gap: There is a lack of features promoting ethical shopping, such as eco-friendly flag.
- ✓ Solution: Introduce user-driven ethical tags to enable buyers to filter and choose products based on their values.

3. AI-Driven Personalization and customization:

- ✓ Gap: Current platforms have a Lack of customization tools in areas like customize profiles and personalization tools, especially in areas like handmade products, AI-driven gifting, and Ai to design.
- ✓ Solution: Implement AI-driven tools that suggest gifts, design ideas, and style advice AI based on user preferences and customization tools.

4. Search and Comparison Tools:

- ✓ Gap: Search functionality on most platforms is basic and lacks intuitive tools like photo search and detailed product comparisons.
- ✓ Solution: Develop advanced search tools that allow users to find products through images and provide detailed comparison features to help users make informed decisions.

5. Social Features:

- ✓ Gap: Current e-commerce platforms are missing strong social shopping elements like Wishlist sharing and gift cards.
- ✓ Solution: Integrate social commerce features, allowing users to share Wish lists, chat, upload reels for 'sellers', and send gift cards.

6. User Engagement and Retention:

- ✓ Gap: Platforms do not provide enough features that encourage repeated user engagement, such as points systems, trend section, or event reminders.
- ✓ Solution: Introduce a point system to reward purchases and activity, provide event reminders for special events, and make a trend section to keep users engaged.
- ✓

7. Handmade and Unique Offerings:

- ✓ Gap: Platforms are not focusing on handmade or unique product sections.
- ✓ Solution: Create a specialized handmade section and enable sellers to post and offer unique items.

2.5 User Needs

- **Visual Search Capability:** Customers want to find products quickly and easily through photo searches, especially for fashion, home decor, and unique items.
- **Personalized Shopping Experience:** Customers expect recommendations that align with their preferences, past purchases, and style. They also want personalized gifting suggestions, and style advice AI.
- **Ethical and Eco-Friendly Shopping:** Many customers prioritize ethical shopping, seeking eco-friendly products.
- **Comparison tool and affordable Alternatives:**
 - ✓ Customers want the ability to compare products side by side to make informed purchasing decisions based on price, features, and reviews.

Price-conscious customers want to find cheaper alternatives to the products they are interested in.
- **Faster Delivery Options:** Some customers need fast shipping, especially for urgent purchases or special occasions.
- **Interactive and Social Features:** Customers enjoy social shopping experiences, such as sharing a Wishlist, sending gift cards, and receiving event reminders.
- **Rewards:** Customers prefer programs that reward them with points for purchases and activities, which can be changed for discounts or special offers.

2.6 Handcrafter Needs

- **Lower Fees:** Sellers want to maximize their profits, especially compared to other e-commerce platforms with higher fees.
- **Customizable Seller Profiles:** Sellers want to create a unique brand identity on the platform, with the ability to customize their profiles and showcase their products effectively.
- **Promotional Tools:** Sellers want features like discount posting, special offers and one-piece discounts to increase visibility and sales.
- **reels:** Sellers want to showcase their products with video content and engage customers.
- **Notifications:** Sellers want notifications about user activity, such as when a customer interacts with their products or posts.
- **Bazar and Offline Ads:** Sellers need offline bazar ads to promote their products to a local audience in addition to online sales.

2.7 Feasibility

➤ Technical Feasibility:

- ✓ **Development Platform:** The Application can be developed using Flutter for cross-platform mobile apps or Kotlin for Native Android apps.
- ✓ **AI Features:** AI-powered features like photo search, gifting recommendation, and style advice AI will require integration with machine learning libraries or platforms. Some of them may need a custom model trained using our data.
- ✓ **Eco-friendly tags:** Each item will have its respective flag stored in its meta-data.
- ✓ **Custom Handmade Products Section:** These can be natively developed into the application or be integrated into the main app as a service or a module.
- ✓ **Database:** NoSQL databases like Fire store or MongoDB handle large volumes of product data efficiently.
- ✓ **Loyalty Points and Gift Cards:** Existing loyalty and gift card APIs can be integrated or custom-built, depending on the needs of the software.
- ✓ **Online Bazars:** This feature can be supported using tools like virtual event management APIs or integrating third-party services.
- ✓ **Comparison Tool:** It can be implemented with search algorithms and price scraping tools, fetching the best results across our product inventory.
- ✓ **Server Hosting:** Hosting on cloud platforms like AWS, Azure, or Google Cloud will offer reliable and scalable servers.

➤ Economic Feasibility:

- ✓ **Initial Development Costs:**
 - **Marketing:** Digital marketing budget for initial user acquisition.
 - **AI integration (Photo search, gift recommendation, style advice):** Data collection and model training costs.
 - **Backend infrastructure:** server and hardware costs.
- ✓ **Operational Costs:**
 - **Cloud hosting services:** Pay-as-you-go models (AWS, Azure) based on traffic.
 - **AI API usage:** Ongoing costs for services like Google Cloud Vision API, or hosting custom-trained models.
 - **Maintenance:** Regular updates, bug fixes, and enhancements will incur ongoing expenses.
 - **Logistics and express delivery:** The cost of the drivers or the third-party logistics company we are contracted with.
 - **Marketing:** The costs of social media ads and deals with popular brands.

✓ **Revenue Streams:**

- **Sales commissions:** Earnings from transactions.
- **Subscription models:** For premium users or businesses.
- **Advertising:** Brands can pay for visibility in trend sections or recommendations.
- **Loyalty Points:** Encouraging user retention, thus increasing long-term revenue.

➤ **Operational Feasibility:**

✓ **Target Audience:**

- **Sellers:** Artisans, small business owners, and hobbyists who produce handmade products.
- **Buyers:** Individuals seeking custom or handmade goods, often focused on eco-friendliness or supporting local crafts.

✓ **User experience:**

- The app must have a smooth and intuitive UI/UX to handle complex features like AI-driven recommendations, wholesale purchases, and event reminders.

✓ **Logistics and Delivery:**

- Integrating with multiple delivery partners for express options will ensure smooth operations and 24/7 express deliveries.

2.8 Risk Analysis

➤ Technical Risks

Risk	Occur	Affect	Solution
Issues with image reorganization	M	H	-Use reliable AI image recognition APIs. - test extensively on a variety of images.
Inappropriate Recommendation	L	H	-Continuously refine AI algorithms. - collect user feedback.
Synchronization issues	L	H	-strong backend scheduling systems. - notification testing.
Inability predicts trends	H	M	-real-time analytics integration.

Table 3: Technical Risks

➤ Resource Risks

Risk	Occur	Affect	Solution
Difficulty of management different types of users	M	H	- Clear responsibility and duties for each type of users.
Delay in Delivery (Low experience – Low Budget)	L	H	- Choose a trusted company to work with. - Manage our delivery (Hard) FP.
Quality Control problem	H	H	- Implement a verification process. - Fair return policy.

Table 4: Resource Risks

➤ **Time Risks**

Risk	Occur	Affect	Solution
A lot of trials for AI systems	H	M	-Use a reliable third-party AI system - Built AI systems from the beginning.

Table 5: Time Risks

➤ **Stakeholders Risk**

Risk	Occur	Affect	Solution
Fake or biased rating	H	H	- Add conditions to allow rating.
Unknown quality on second-hand and handmade	H	H	- Make customers choose materials, sizes...
Inconsistent user experience on Bazar	M	H	- Make every seller clarify all things about products.
Misleading eco-friendly flags	M	M	- Ensure transparent verification processes. - Clarify criteria for applying such flags to products.

Table 6: Stakeholders Risk

Chapter 3: San3a Analysis

3.1 Functional Requirements.

1. Search by Photo

- **Description:** Users can upload a photo of a product they like, and the platform uses image recognition technology to find similar or identical products available on the website.
- **Benefit:** Simplifies the search process for users who may not know the name or description of the product but have a visual reference.

2. Chat

- **Description:** A direct messaging system that allows customers to communicate with artisans or sellers to discuss product details, customization options, or any queries.
- **Benefit:** Enhances customer satisfaction by enabling personalized communication and building trust between buyers and sellers.

3. Custom Handmade Products Section

- **Description:** Users can upload a description or design of a custom handmade product they need. Artisans or sellers can then contact the user to discuss and fulfill the request.
- **Benefit:** Encourages creativity and personalization, catering to users looking for unique, one-of-a-kind items.

4. AI-Driven Gifting Feature

- **Description:** An AI system that recommends gift ideas based on user-provided information such as the recipient's age, gender, interests, and occasion.
- **Benefit:** Saves time for users who struggle to find the perfect gift and increases sales by suggesting relevant products.

5. Event Reminder System

- **Description:** Users can input important dates (e.g., birthdays, anniversaries) into the platform, and the system sends reminders before the event, along with gift suggestions.
- **Benefit:** Helps users stay organized and ensures they never miss an important occasion.

6. Comparison Tool

- **Description:** Users can select Two products, and the platform compares them based on criteria like price, material, brand, and customer reviews.
- **Benefit:** Helps users make informed purchasing decisions by providing a side-by-side analysis of products.

7. Rating System

- **Description:** Each product has a rating and reviews from users who have purchased and used it.
- **Benefit:** Builds trust and credibility, helping users choose high-quality products based on peer feedback.

8. Trend Section

- **Description:** A dedicated section showcasing the best-selling or most popular products on the platform.
- **Benefit:** Keeps users updated on current trends and encourages them to explore popular items.

9. Recommendation System

- **Description:** An AI-powered system that suggests products based on the user's browsing history, preferences, and past purchases.
- **Benefit:** Enhances the shopping experience by offering personalized product recommendations

10. AI to Design

- **Description:** Users can describe their desired design (e.g., color, style, material) to an AI chatbot, which generates a visual representation of the product.
- **Benefit:** Simplifies the design process for users and helps them visualize their custom orders.

11. Create Wishlist

- **Description:** Users can save products they are interested in a Wishlist for future reference or purchase.
- **Benefit:** Helps users keep track of items they want to buy later.

12. Share Wishlist

- **Description:** Users can share their Wishlist with friends or family via a link or email.
- **Benefit:** Makes it easier for others to buy gifts for the user based on their preferences.

13. Gift Cards

- **Description:** Users can purchase and send digital or physical gift cards that recipients can use to buy products on the platform.
- **Benefit:** Provides a flexible gifting option and encourages repeat purchases.

14. Points System

- **Description:** Users earn points for every purchase, which can be redeemed for discounts or free products in the future.
- **Benefit:** Encourages customer loyalty and repeat purchases.

15. Special Order (Faster Delivery)

- **Description:** Users can place special orders for faster delivery, often at an additional cost.
- **Benefit:** Provides flexibility for users who need products urgently.

16. History of Watched Products

- **Description:** A log of products the user has viewed or interacted with in the past.
- **Benefit:** Allows users to easily revisit products they are interested in.

17. Ads for Offline Bazar

- **Description:** The platform displays advertisements for offline bazaars, workshops, or events related to handmade products.
- **Benefit:** Promotes local businesses and encourages community engagement.

18. Bazar on Application

- **Description:** A virtual bazaar where sellers offer products at discounted prices for a limited time.
- **Benefit:** Creates a sense of urgency and excitement, driving sales for sellers.

19. Add Order Post

- **Description:** Customers can create a post detailing their desired product, including specifications, budget, and timeline.

- **Benefit:** Allows sellers to bid on or fulfill custom orders, catering to specific customer needs.

20. Add Offers on Posts

- **Description:** Artisans or sellers can respond to customer posts with offers, including pricing and delivery timelines.
- **Benefit:** Facilitates a competitive marketplace and ensures customers get the best deal.

21. Notifications for Sellers

- **Description:** Sellers receive notifications when a new post or order is added that matches their expertise or product category.
- **Benefit:** Ensures sellers don't miss potential business opportunities.

3.2 Non-Functional Requirements.

Requirement	Details
Performance	The app should load product listings within 10 seconds for 90% of the users The AI-driven features should provide suggestions within 30 seconds of user input.
Scalability	The application should handle up to 1,000 concurrent users without performance degradation.
Availability	The app should be available 90% of the time, with downtime limited to maintenance windows.
Security	The system must be protected from unauthorized access: the user must enter an email address and password to log into the account All sensitive data must be encrypted using industry-standard encryption methods Must ensure secure transmission of payment information during payment processing
Compatibility	The app must be compatible with Android devices that run OS version 10 and above
Localization	The app must support both Arabic and English, allowing users to switch languages easily
Reliability	The system should be working without any failures for 90% of the month. The app should be able to recover from minor faults without crashing
Usability	The app should follow consistent design language and UI guidelines All text should comply with WCAG AA for large and normal text

Table 7: Non-Functional Requirements

3.3 User Persona

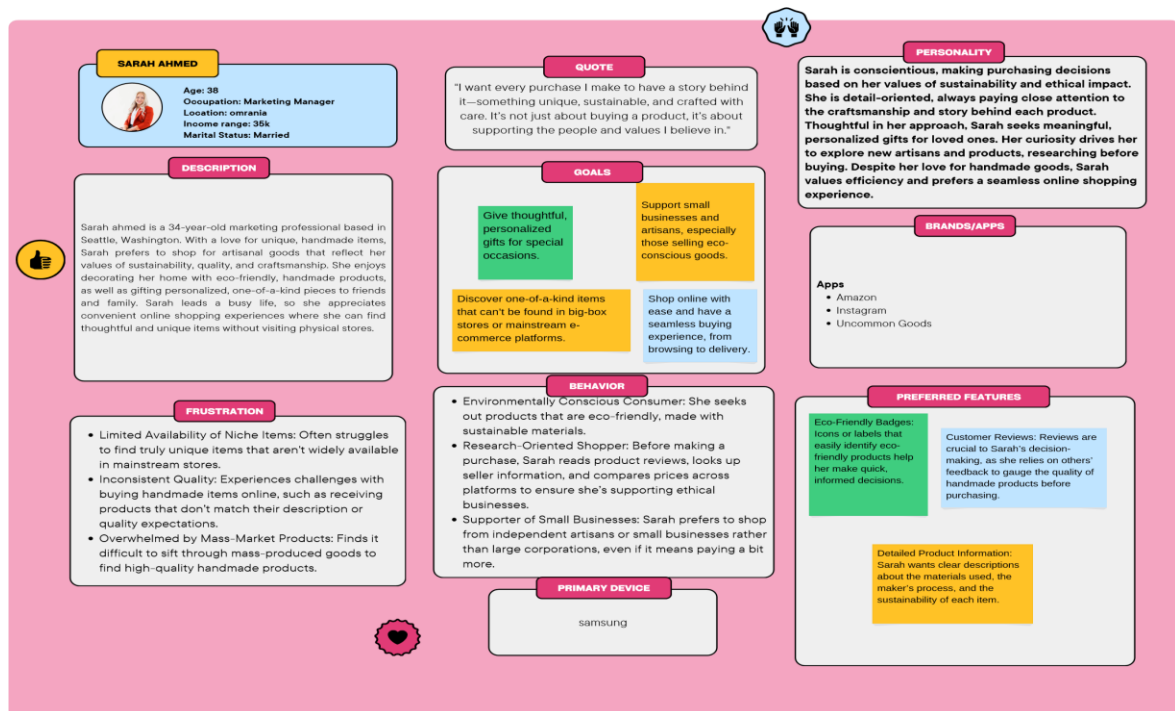


Figure 13: User Persona 1

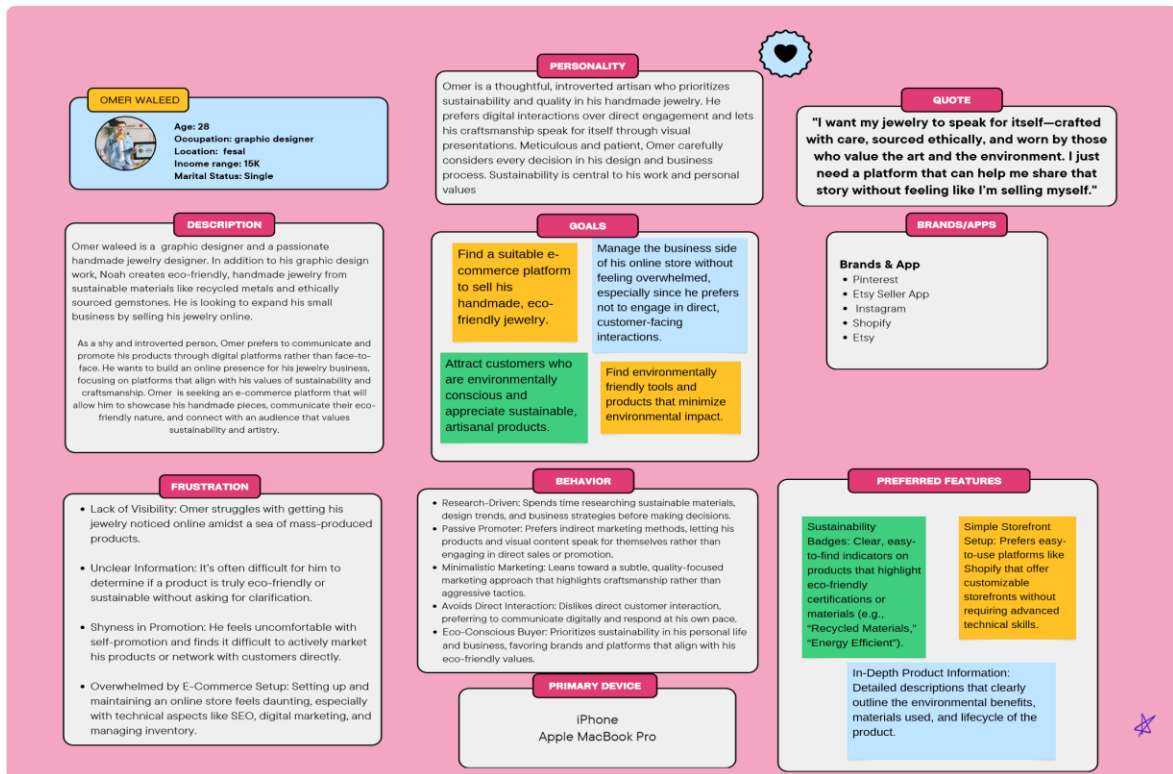


Figure 14: User Persona 2

3.4 Use Case



Figure 15: Use Case

3.5 User Stories

1. Search by Photo

User Story name	Search by photo.	
Scenario	A customer wants to search for a product by photo.	
Triggering event	The customer presses the camera icon on the search bar.	
Brief description	Customer wants to upload an image to find this product on application so that he does not need to write the product name to find it.	
Actors	Customer	
Preconditions	None	
Postconditions	Go to the products list page	
Flow of activities	Actor	System
	1- Customer presses the camera icon on the search bar.	
		2- System presents options: upload an image or take a photo.
	3- Customer selects an option and provides an image (takes a photo or uploads from the gallery).	
		4- The system goes to the products list page.
Exception conditions	No Match Found	

Table 8: Search By Photo User Story

2. Manage Wishlist

User Story name	Manage Wishlist	
Scenario	A customer wants to add, delete, and send his Wishlist.	
Triggering event	The customer goes to edit his Wishlist.	
Brief description	The customer wants to delete or edit an existing item or wants to share his Wishlist with another one so that he can tell him what he wants to buy without the need to send each product individually	
Actors	Customer	
Preconditions	Create Wishlist	
Postconditions	The system performs the order	
Flow of activities	Actor	System
	1. Customer opens the Wishlist section.	
		2. System displays the customer's current Wishlist items.
	3. Customer chooses to edit an existing item or share the Wishlist.	
		4. System executes the order
Exception conditions		

Table 9: Manage Wishlist User Story

3. Ask for Gift Recommendation

User Story name	Ask gift recommendations	
Scenario	A customer wants to have a recommendation about gifts.	
Triggering event	The customer goes to the recommendation page.	
Brief description	When a customer is seeking gift ideas, they can enter person information so that the application can recommend list of products to him.	
Actors	Customer	
Preconditions	None	
Postconditions	View products	
Flow of activities	Actor	System
	1. Customer opens the gift recommendations page.	
		2. System get the person preferences from him.
	3. Customer answers all the questions.	
		4. System displays more than one product according to his answers.
Exception conditions	None	

Table 10: Ask For Gift Recommendation User Story

4. Add Event Reminder

User Story name	Add event reminder	
Scenario	A customer wants to add a reminder for the event.	
Triggering event	The customer goes to the reminders page.	
Brief description	When a customer has an important event like birthday or anything else, he can add details (date, name, location) to the application and the application sends notification to him before the event.	
Actors	Customer	
Preconditions	None	
Postconditions	System send notification on the time	
Flow of activities	Actor	System
	1. Customer Go to reminders page.	
	2. Customer presses on add new reminder.	
		3. System requests event information from the user.
	4. Customer enters event information	
		5. System sends a notification for the customer on the time.
Exception conditions	None	

Table 11: Add Event Reminder User Story

5. Comparing Products

User Story name	Compare products	
Scenario	Customer wants to compare between two products	
Triggering event	Customer presses on compare products icon	
Brief description	Customer wants to compare between two items on the application according to (price, material, seller rate) so that he can choose the best product for him	
Actors	Customer	
Preconditions	None	
Postconditions	None	
Flow of activities	Actor	System
	1- Customer goes to comparison page	
		2- The system allows the user to enter the name of the products
	3- Customer enters each product name	
		4- System displays all products that user enters its names
	5- Customer chooses the products	
		6- System Display the final comparison
Exception conditions	Product does not exist	

Table 12: Comparing Products' User Story

6. Generate Design

User Story name	Generate design	
Scenario	Customer wants to generate design	
Triggering event	Customer goes to chat bot	
Brief description	Customer wants to generate an image for any product according to some instructions that he gives to the bot so that he can have an image that he can show it to seller or make a post that someone can make it for him	
Actors	Customer	
Preconditions	None	
Postconditions	Make post	
Flow of activities	Actor	System
	1- Customer goes to chat bot	
	2- Customer writes instruction	
		3- The system sends responses with generated images
	4- Customer confirms the image or repeat step two	
	5- Customer creates order post	
		6- System reviews the post
Exception conditions	Post rejected	

Table 13: Generate Design User Story

7. Buying Gift Card

User Story name	Buy gift card	
Scenario	The customer wants to buy gift card	
Triggering event	The customer goes to buy gift card page	
Brief description	Customer wants to buy gift card to send it to someone or he wants to save it for later use so that he can surprise someone and send it as a gift	
Actors	Customer	
Preconditions	Customer login	
Postconditions	Check out	
Flow of activities	Actor	System
	1- Customer goes to buy gift card page	
		2- The system takes all needed details from customer
	3- The customer goes to check out the screen	
		4- System confirms the order and add the card on user account
Exception conditions	The check out process failed	

Table 14: Buying Gift Card User Story

8. Create Post Order

User Story name	Create post order	
Scenario	The customer wants to buy customized piece	
Triggering event	The customer goes to create post	
Brief description	Customer has design and he go to add the post details (product name, description, image, budget, time) so that he can find someone can make the product for him	
Actors	Customer	
Preconditions	Customer login	
Postconditions	None	
Flow of activities	Actor	System
	1- Customer goes to create post	
		2- System shows the post form for the customer to complete it and choose all options
	3- Customer chooses the options and complete the form	
		4- System reviews the post
Exception conditions	The post rejected	

Table 15: Create Post Order User Story

9. Make Normal Order

User Story name	Make normal order	
Scenario	The customer wants to buy products	
Triggering event	The customer chooses products	
Brief description	Customer saw products that he like, and he add it to cart so that he can buy it after he chooses the rest items	
Actors	Customer	
Preconditions	Customer login	
Postconditions	checkout	
Flow of activities	Actor	System
	1- Customer chooses products	
		2- System adds the chosen products to the cart
	3- Customer ends shopping and checkout	
		4- System confirms the order
Exception conditions	- Check out failed - Customer chooses in demand product	

Table 16: Make Normal Order User Story

10. Make in Demand order

User Story name	Make in demand order	
Scenario	The customer wants to buy in demand product	
Triggering event	The customer chooses in demand product	
Brief description	Customer saw in demand product that he like, and he make an order by this item so that he can buy it	
Actors	Customer	
Preconditions	Customer login	
Postconditions	None	
Flow of activities	Actor	System
	1- Customer chooses in-demand product	
		2- The system displays all information about the product (price, time, ...)
	3- Customer confirms the order	
		4- The system sends the order to the handcrafter
Exception conditions	The customer does not confirm the order	

Table 17: Make in Demand Order User Story

11. Find History of Watch

User Story name	Find history of watch	
Scenario	The customer wants to saw his history of watch	
Triggering event	The customer presses on history of watch icon	
Brief description	Customer wants to go to his history of watch so that he can find something that he saw before	
Actors	Customer	
Preconditions	Customer login	
Postconditions	None	
Flow of activities	Actor	System
	1- Customer goes to history of watch screen	
		2- System displays all products he saw before
Exception conditions	None	

Table 18: Find History of Watch User Story

12. Add Advertisement

User Story name	Add advertisement	
Scenario	The customer wants to add advertisement for workshop or offline bazar on application	
Triggering event	The customer goes to add advertisement option	
Brief description	Customer had workshop or offline bazar and want to advertise about it so that the handcrafters and customers can know about it and attend	
Actors	Customer	
Preconditions	Customer login	
Postconditions	Check out	
Flow of activities	Actor	System
	1- Customer chooses add advertisement	
		2- System displays rules to add advertisement
	3- Customer confirms the rules	
	4- Customer goes to check out screen	
		5- System reviews the advertisement
Exception conditions	- Check out process failed - Advertisement rejected	

Table 19: Add Advertisement User Story

13. Add Product

User Story name	Add Product	
Scenario	The Handcrafter want to add product on application	
Triggering event	The Handcrafter goes to add product page	
Brief description	Handcrafter has an item and want to add it on his page on application so that customers can demand it	
Actors	Handcrafter	
Preconditions	Has Seller account	
Postconditions	None	
Flow of activities	Actor	System
	5- The handcrafter goes to add product page	
		6- The system shows all the necessary details
	7- The handcraft fills all information needed	
	8- the handcraft posts the product.	
		9- System verifies the product 10- system upload the product on handcrafter page
Exception conditions	The product is violating rules	

Table 20: Add Product User Story

Table 21: Create Own Profile User Story

14. Join Bazar

User Story name	Join Bazar	
Scenario	The Handcrafter want to join on an online bazar	
Triggering event	The Handcrafter ask to join bazar	
Brief description	Handcrafters watch an announced bazar and want to join to it to offer his products so that he can be more famous and achieve more revenue	
Actors	Handcrafter	
Preconditions	Has Seller account	
Postconditions	Add products information	
Flow of activities	Actor	System
	1- The handcraft send request to join bazar	
		2- System reviews his profile
		3- System asks him to add his products information
	4- User adds the products that he will join with it	
		5- System adds his products to the bazar
Exception conditions	User request rejected	

Table 22: Join Bazar User Story

15. Add Reels

User Story name	Add reels	
Scenario	The Handcrafter want to add reel on the application	
Triggering event	The Handcrafter go to add reels page	
Brief description	Handcrafters want to add reels on application about his work and how he performs it so that he can attract more customers	
Actors	Handcrafter	
Preconditions	Has Seller account	
Postconditions	None	
Flow of activities	Actor	System
	1- The handcraft goes to add reel	
		2- System opens the page that he will upload his reel
	3- User uploads his video	
		4- System reviews the video then add it on application
Exception conditions	The video rejected	

Table 23: Add Reels User Story

16. Find Analysis

User Story name	Add reels	
Scenario	The Handcrafter want to add reel on the application	
Triggering event	The Handcrafter go to add reels page	
Brief description	Handcrafters want to add reels on application about his work and how he performs it so that he can attract more customers	
Actors	Handcrafter	
Preconditions	Has Seller account	
Postconditions	None	
Flow of activities	Actor	System
	1- The handcraft goes to add reel	
		2- System opens the page that he will upload his reel
	3- User uploads his video	
		4- System reviews the video then add it on application
Exception conditions	The video rejected	

Table 24: Find Analysis User Story

Chapter 4: San3a Design and Implementation

4.1 Entity Relational Diagram (ERD)

- Completed Schema

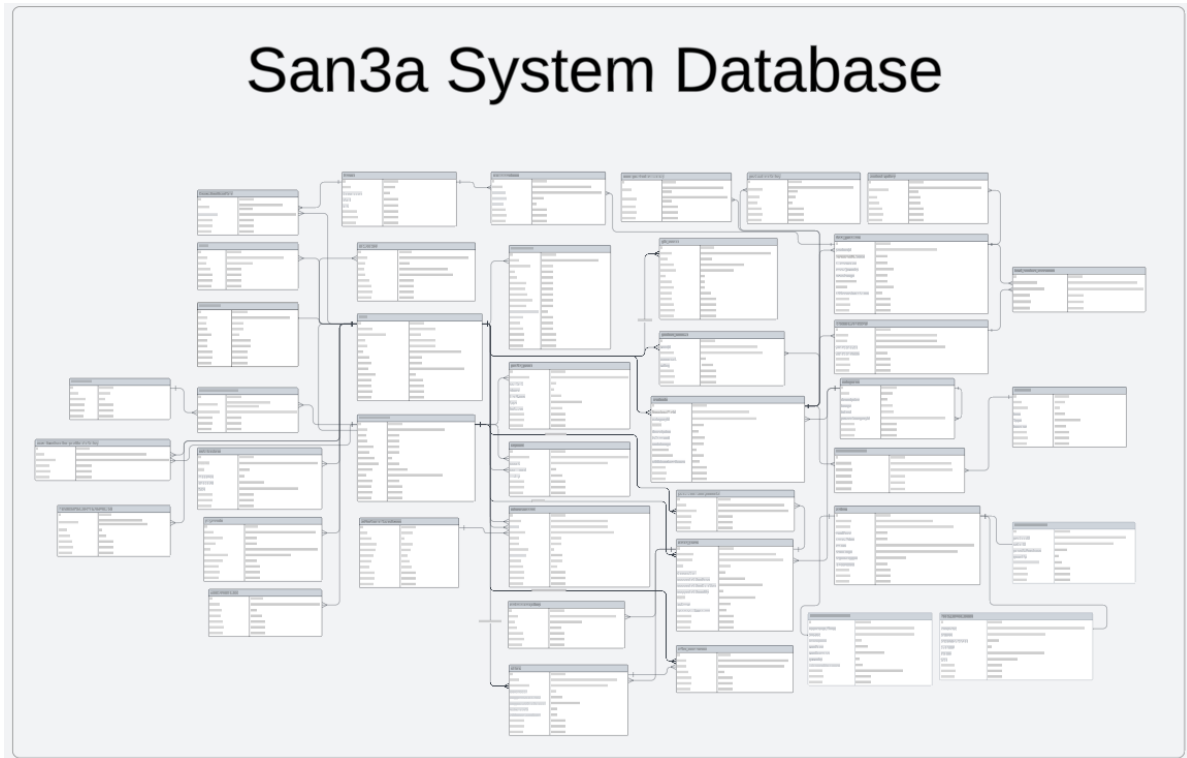


Figure 16: Complete Schema

- Customized Offer Schema

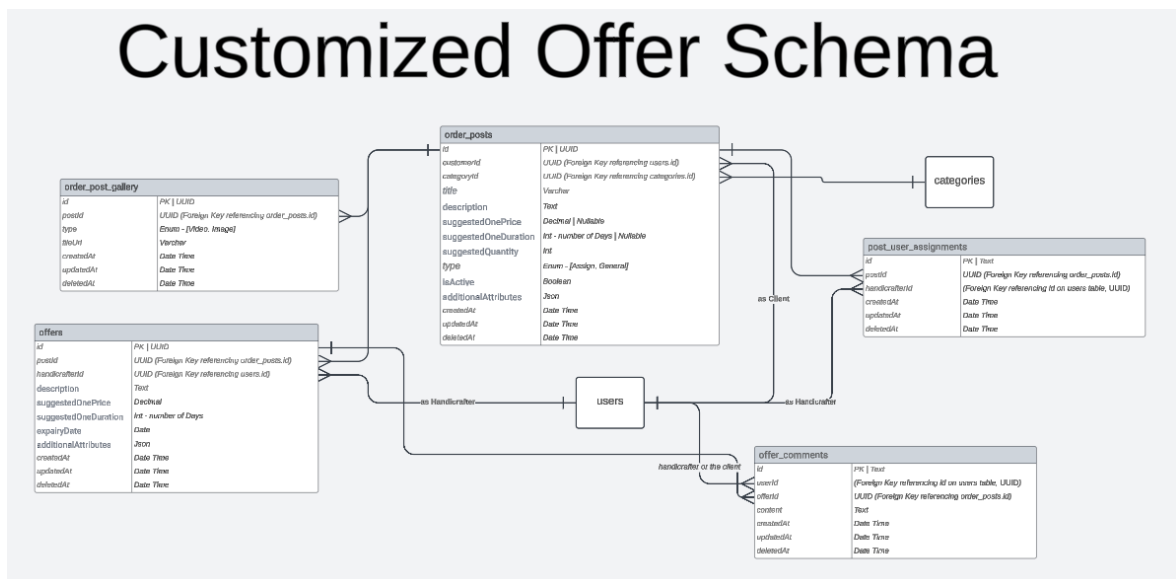


Figure 17: Customized Offer Schema

- User Profiles Schema

Users Profiles Schema

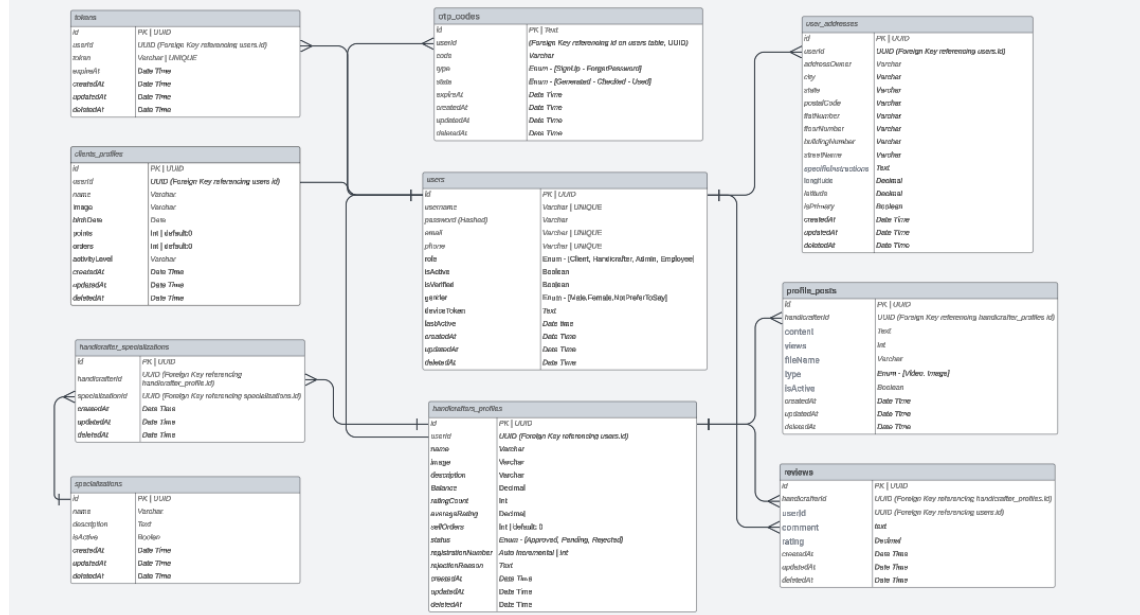


Figure 18: User Profile Schema

- Bazar Schema

Bazars Schema

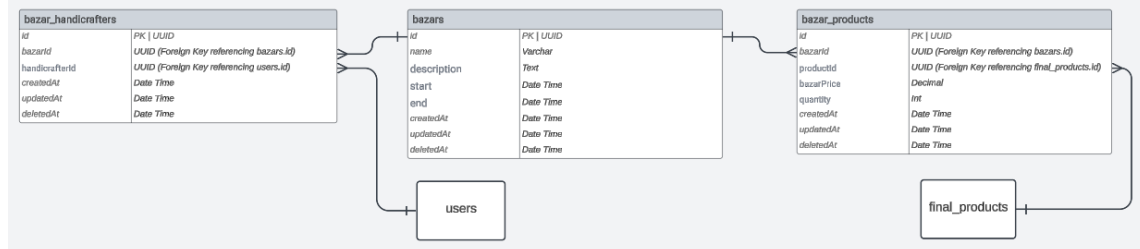


Figure 19: Bazar Schema

- Code And Advertisement Schema

Codes And Advertisement Schema

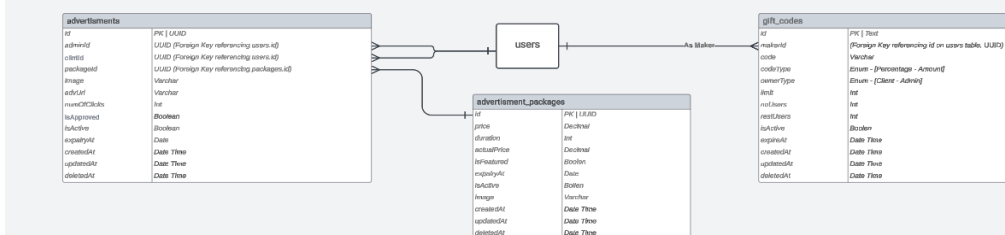


Figure 20: Code and Advertisement Schema

- Order Schema

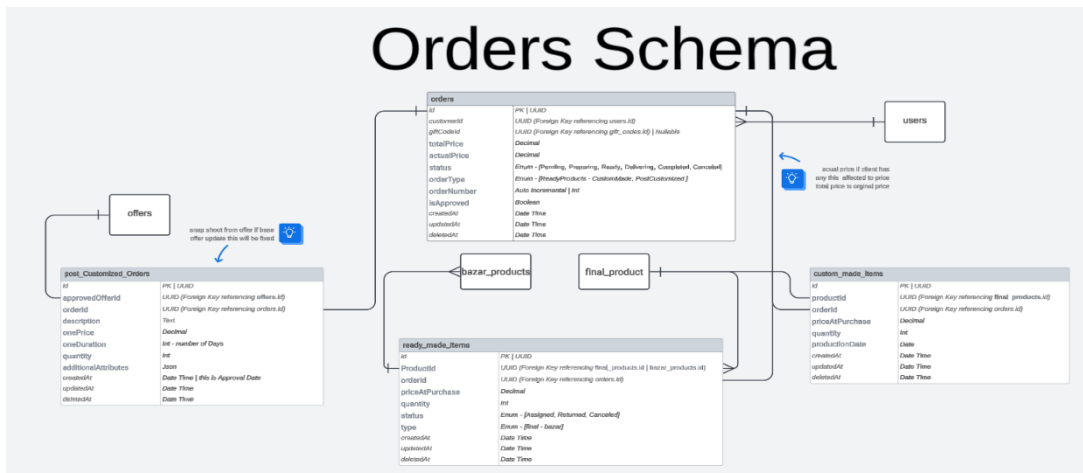


Figure 21: Orders Schema

- Standard Product Schema

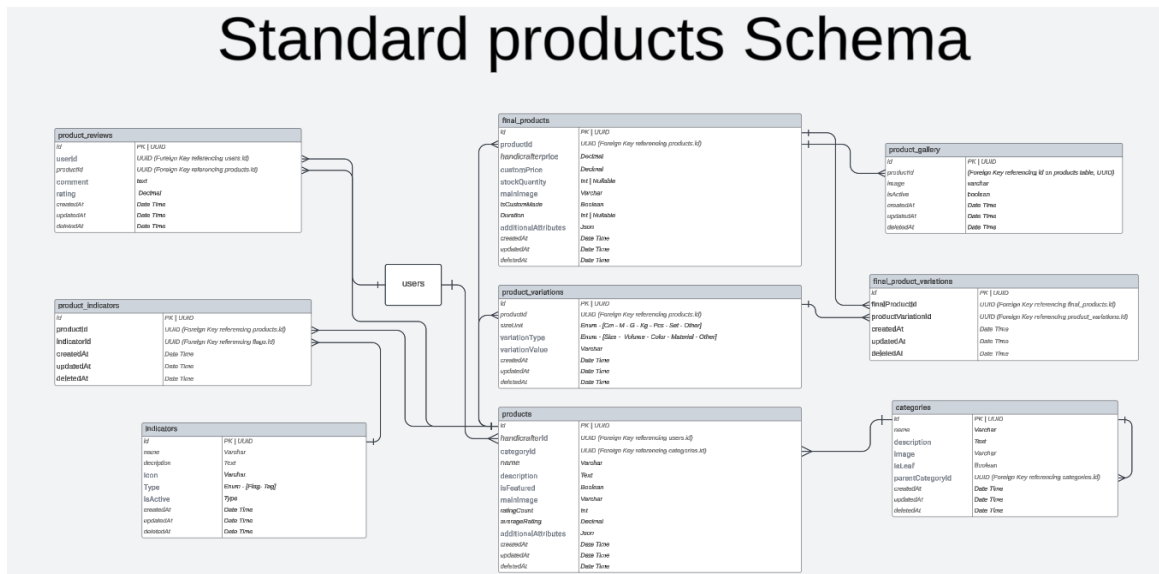


Figure 22: Standard Products Schema

- Logs Schema

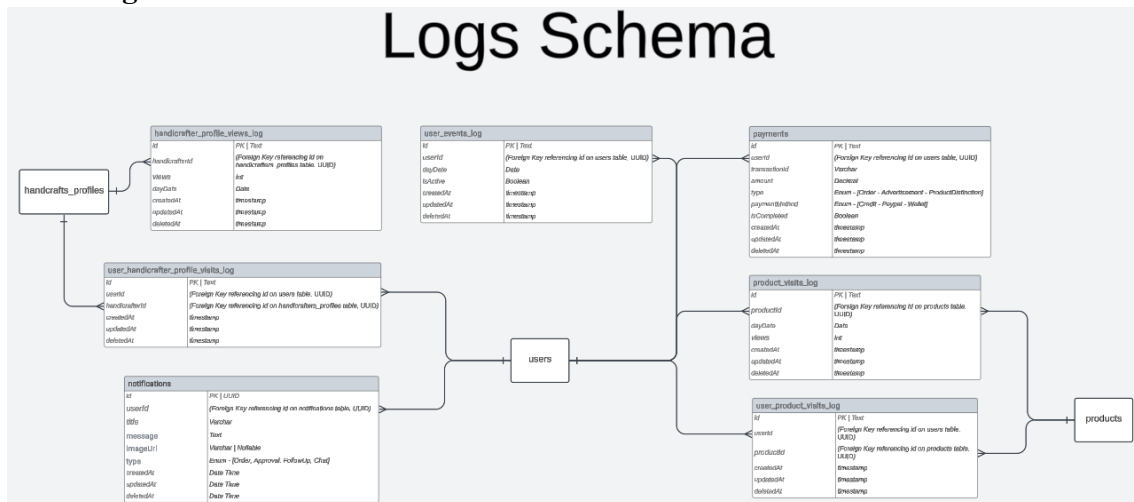


Figure 23: Logs Schema

4.2 Class Diagram

- Complete Class Diagram

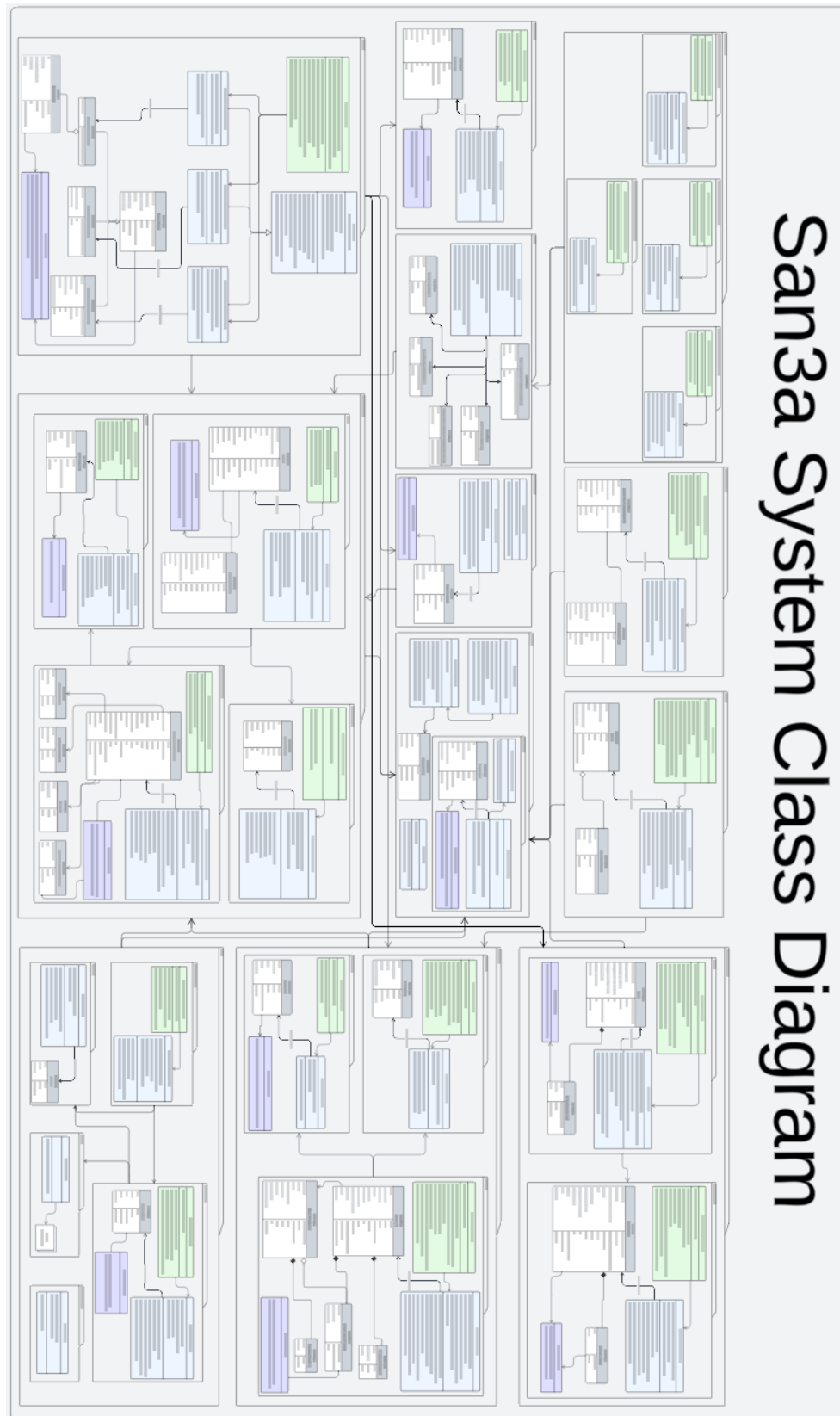


Figure 24: Complete Class Diagram

- Design Principles & Patterns

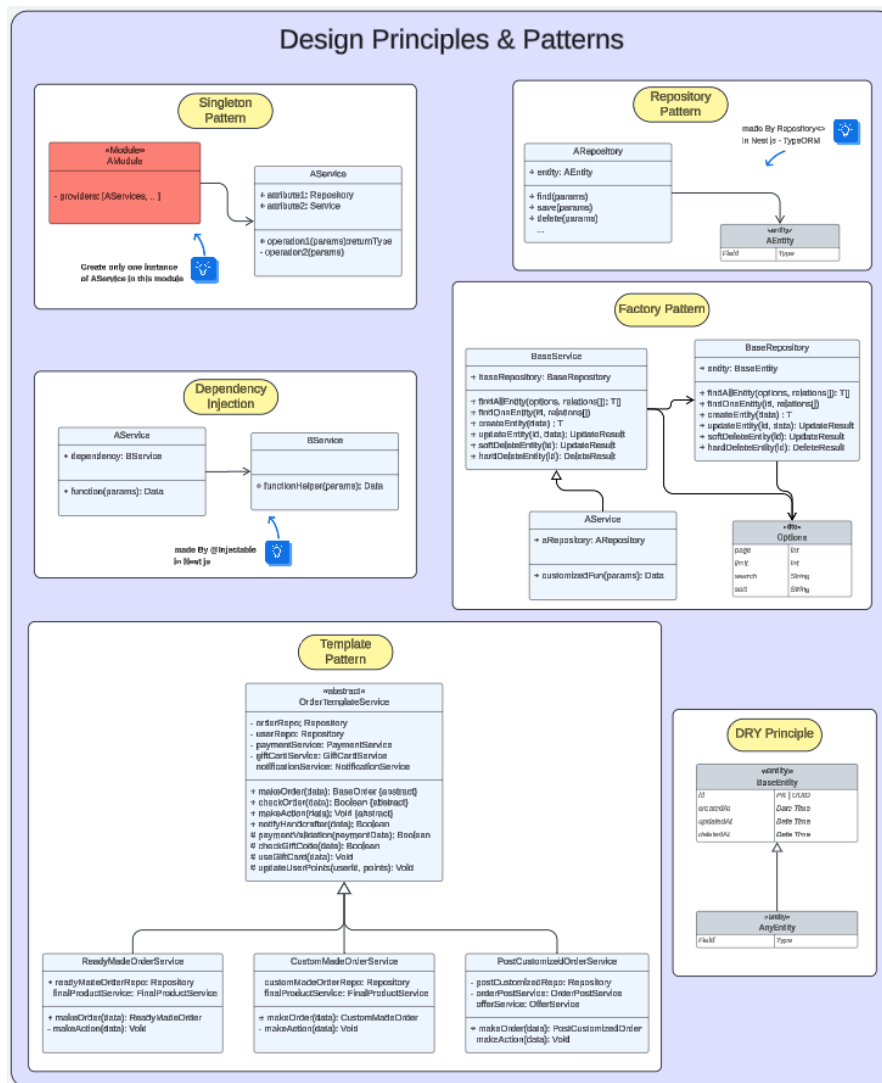


Figure 25: Design Principles & Patterns

- Customized Product

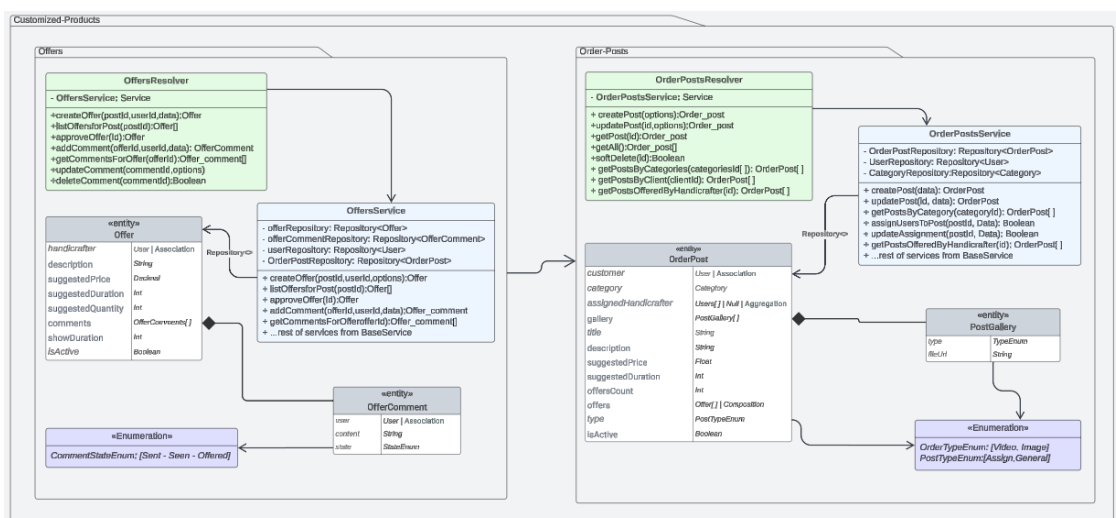


Figure 26: Customized Product Class Diagram

• Products

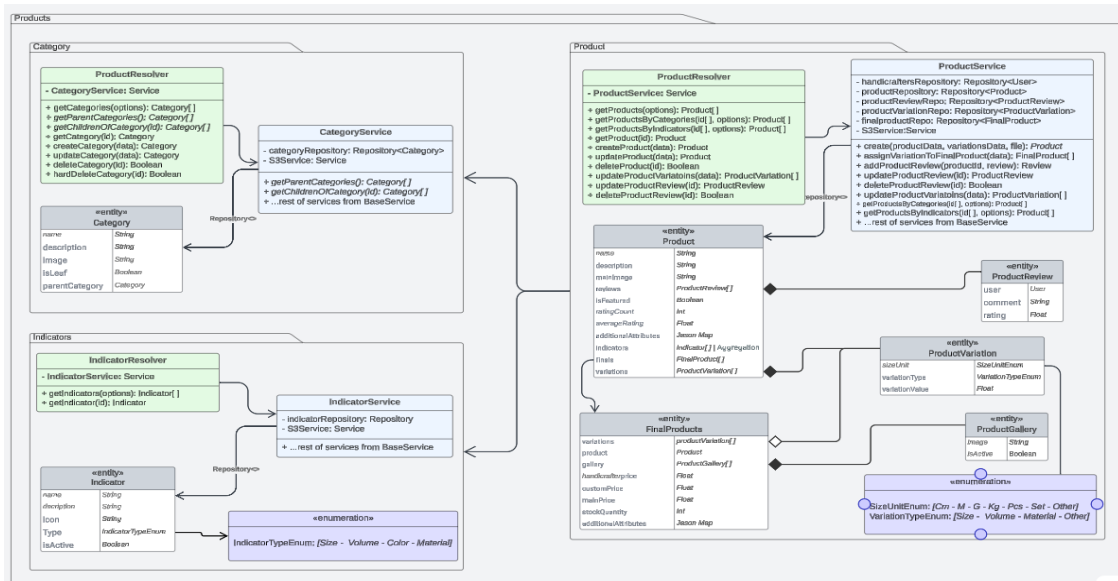


Figure 27: Products Class Diagram

• Authentication

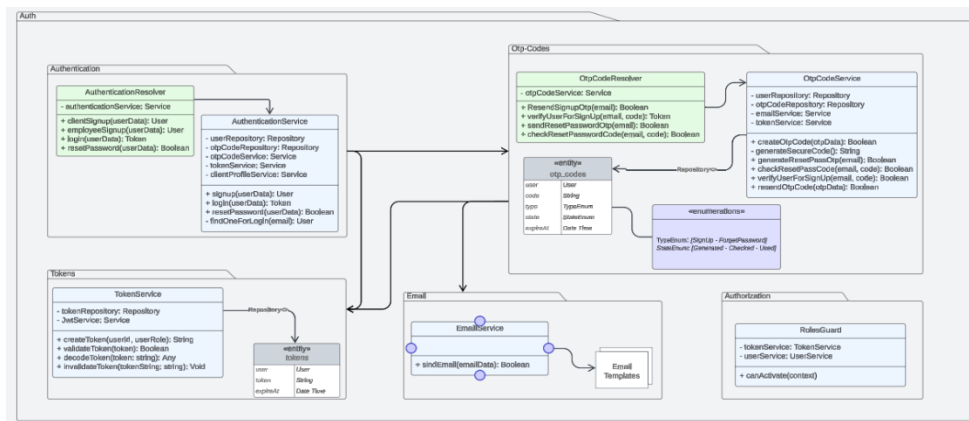


Figure 28: Authentication Class Diagram

• Bazar And Advertisements

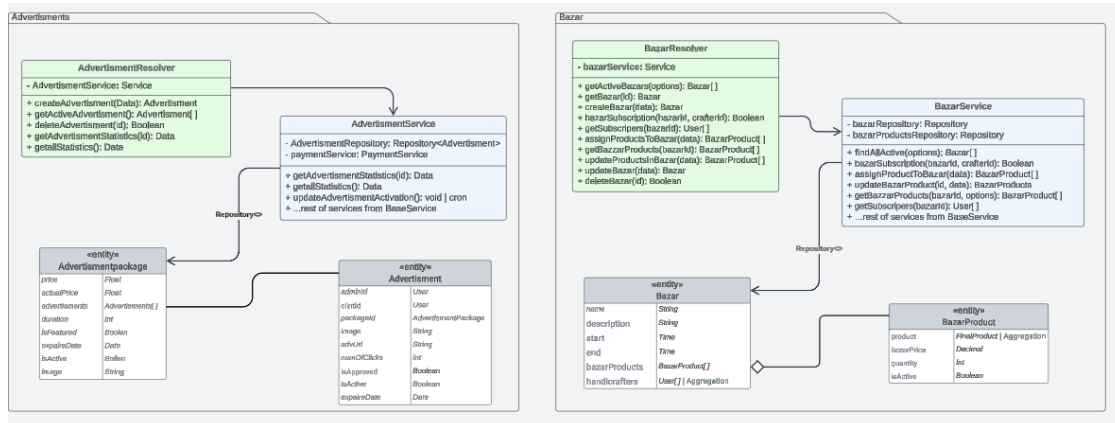


Figure 29: Bazar and Advertisement Class Diagram

• Notifications and shared

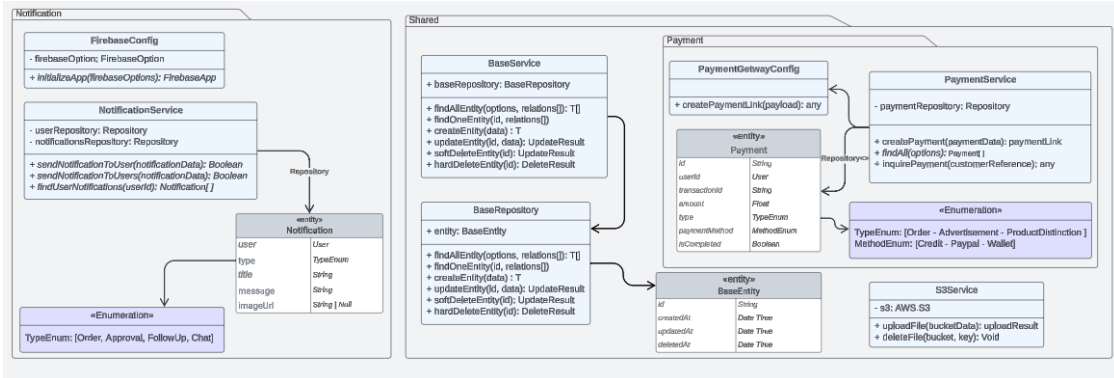


Figure 30: Notification and Shared Class Diagram

• User Profiles

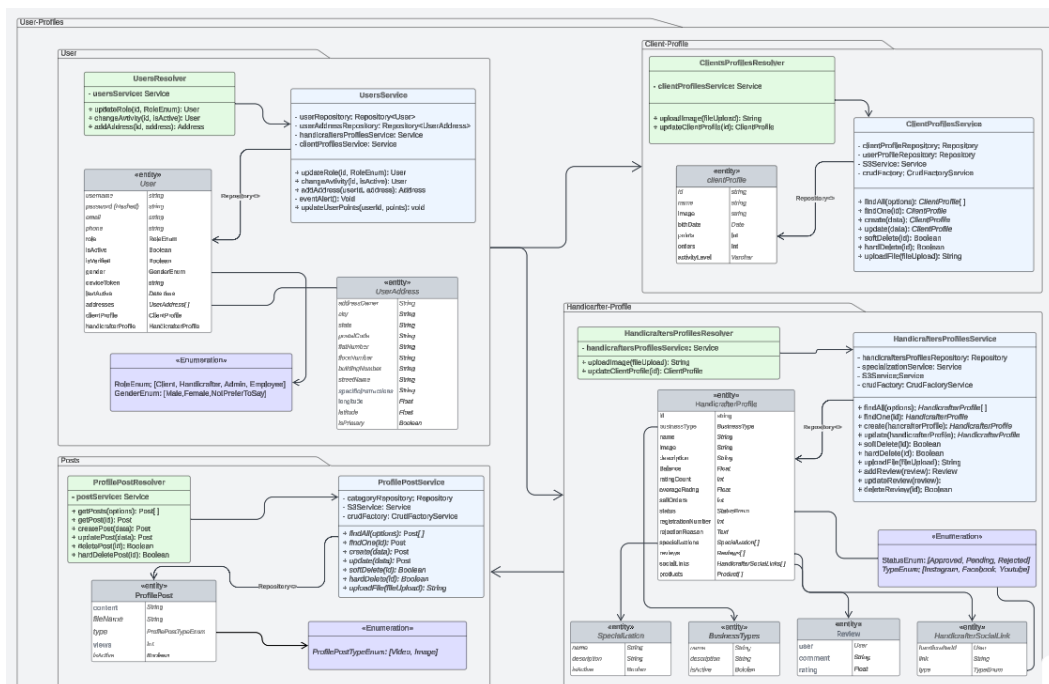


Figure 31: User Profiles Class Diagram

• AI Integrations

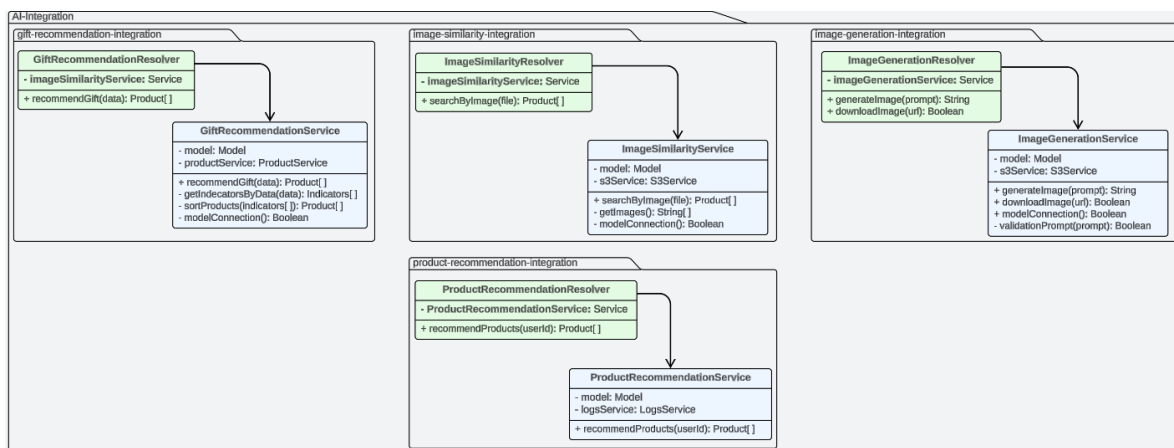


Figure 32: AI Integrations Class Diagram

• Logs and Gift Codes



Figure 33: Logs and Gift Codes Class Diagram

• Orders

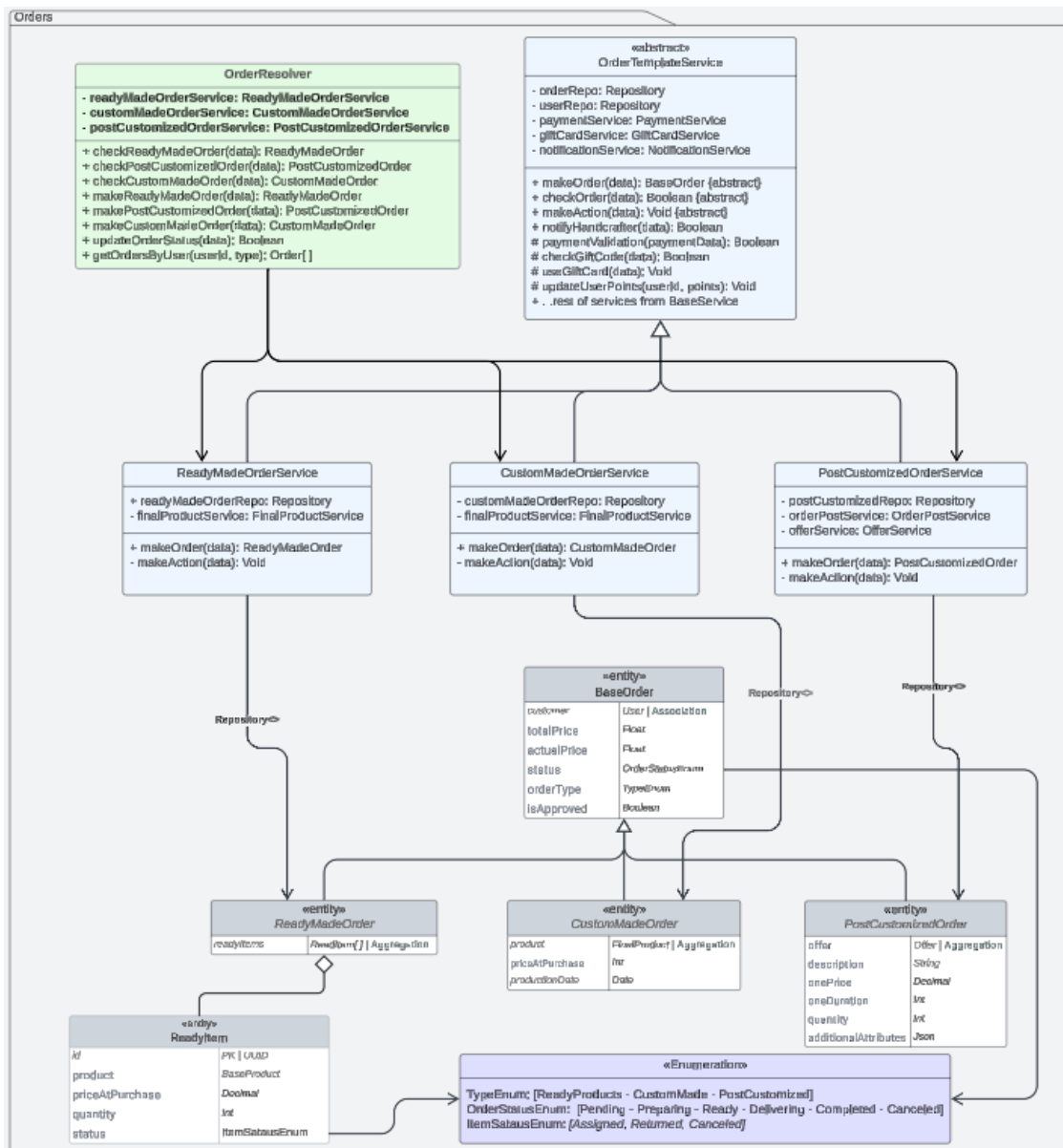


Figure 34: Orders Class Diagram

4.3 Sequence Diagram

4.3.1 Add Advertisement

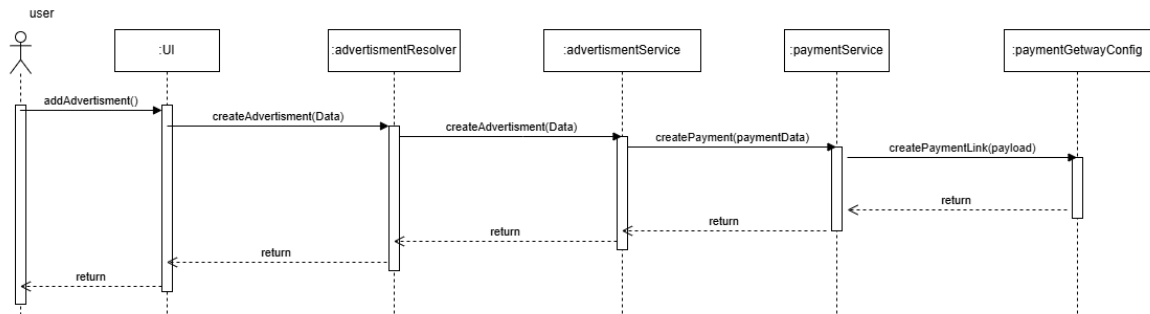


Figure 35: Add Advertisement Sequence Diagram

4.3.2 Add Offer

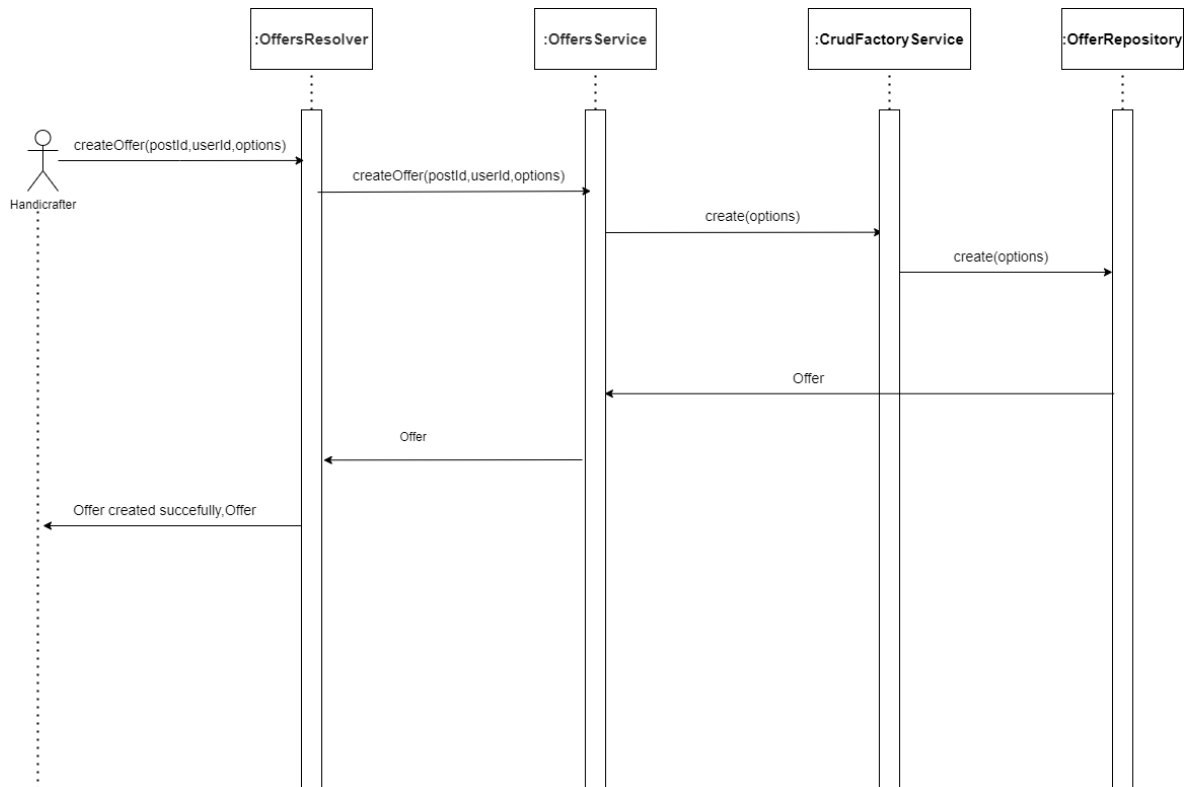


Figure 36: Add Offer Sequence Diagram

4.3.3 Add Post

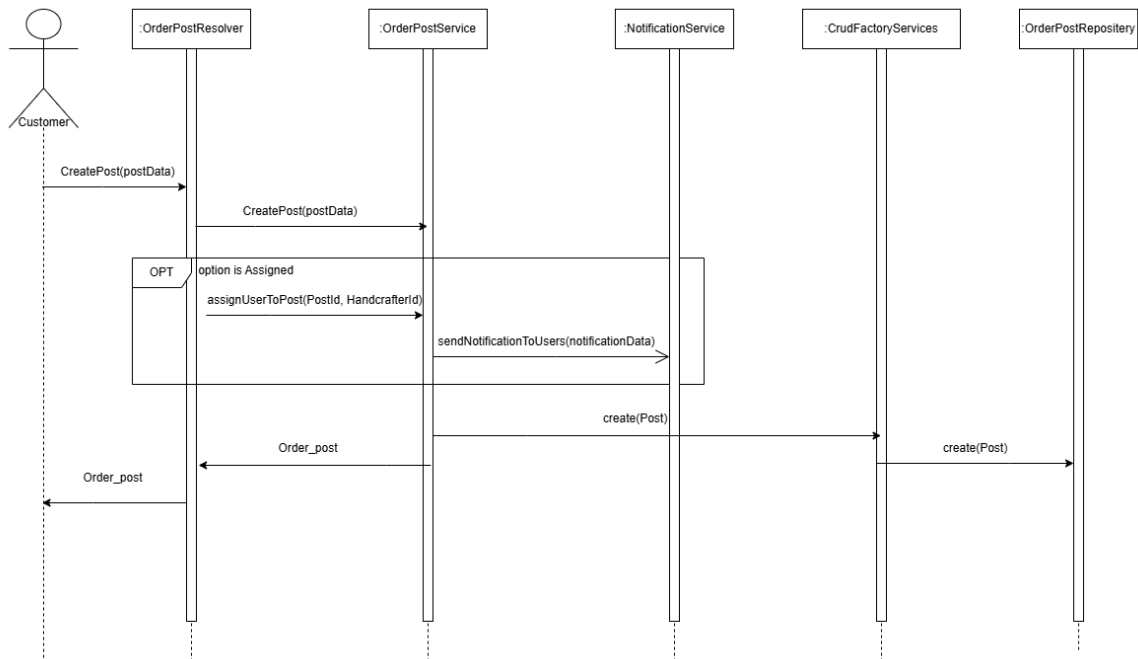


Figure 37: Add Post Sequence Diagram

4.3.4 Add Product

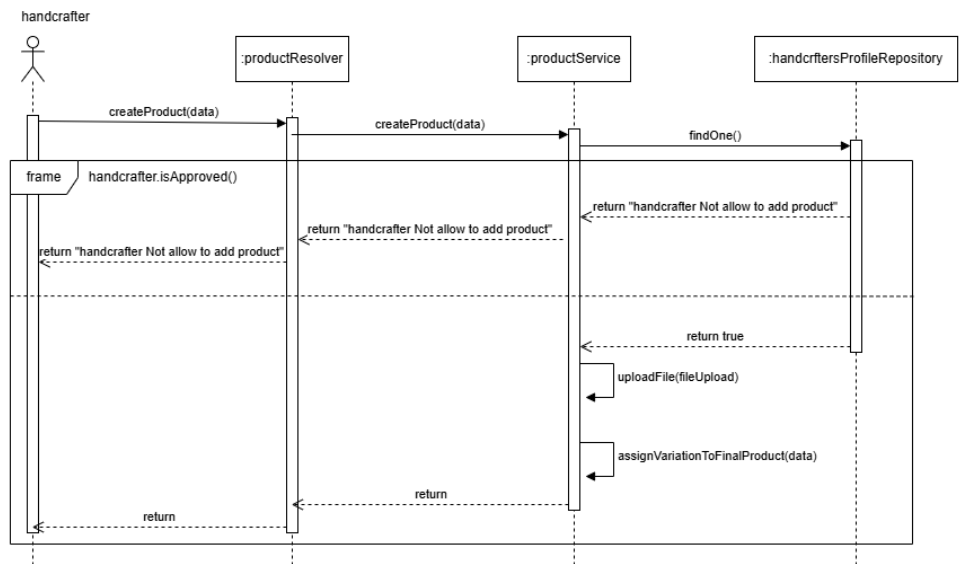


Figure 38: Add Product Sequence Diagram

4.3.5 Add Review

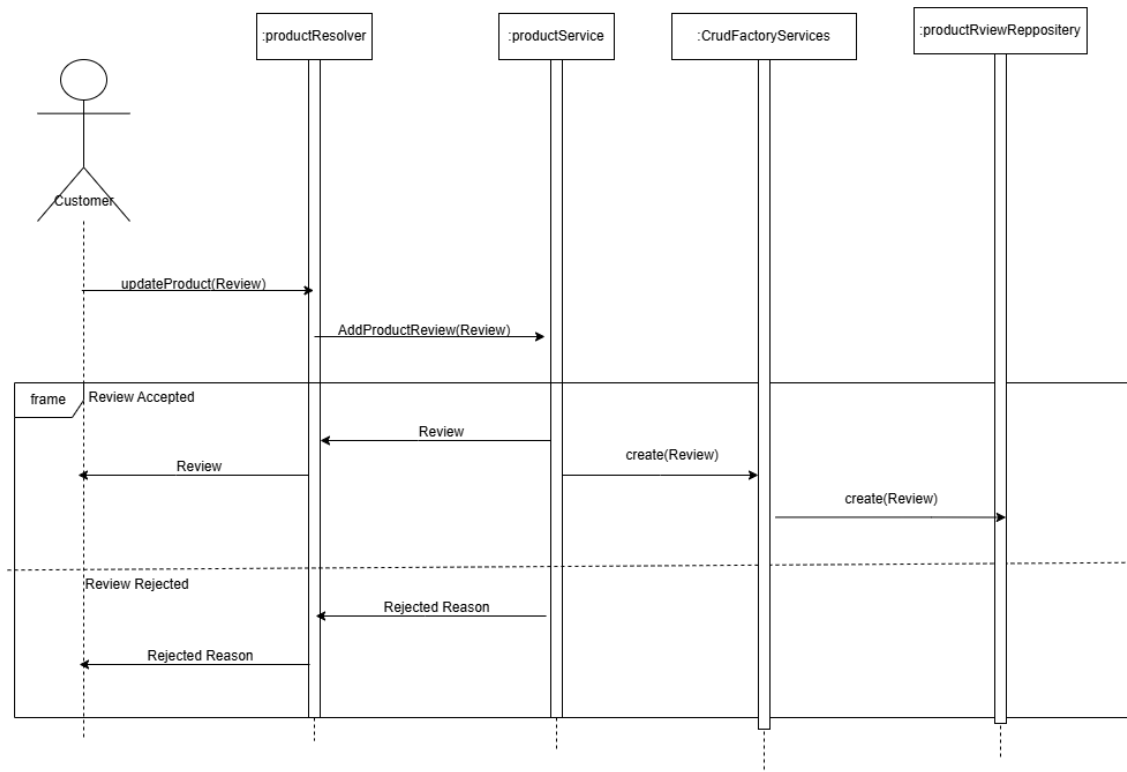


Figure 39: Add Review Sequence Diagram

4.3.6 Approve Offer

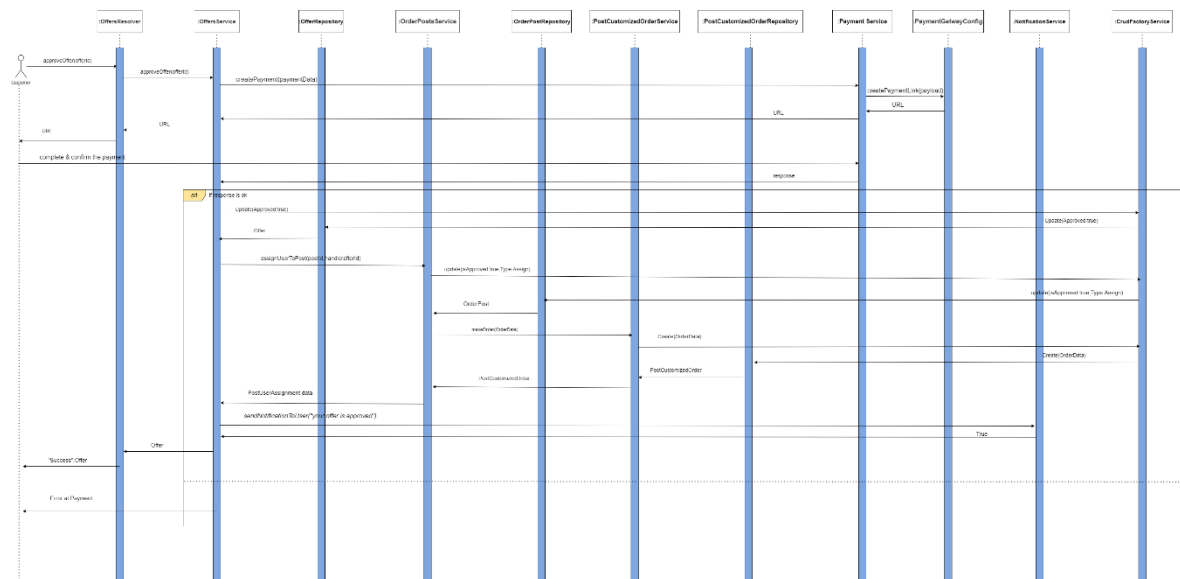


Figure 40: Approve Offer Sequence Diagram

4.3.7 Forget Password

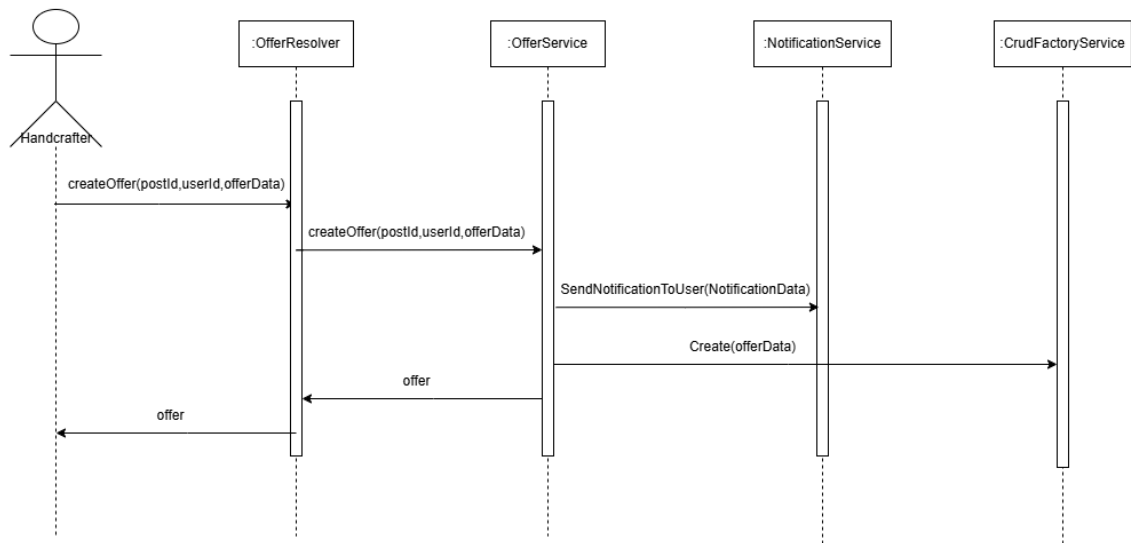


Figure 41: Forget Password Sequence Diagram

4.3.8 Generate Gift

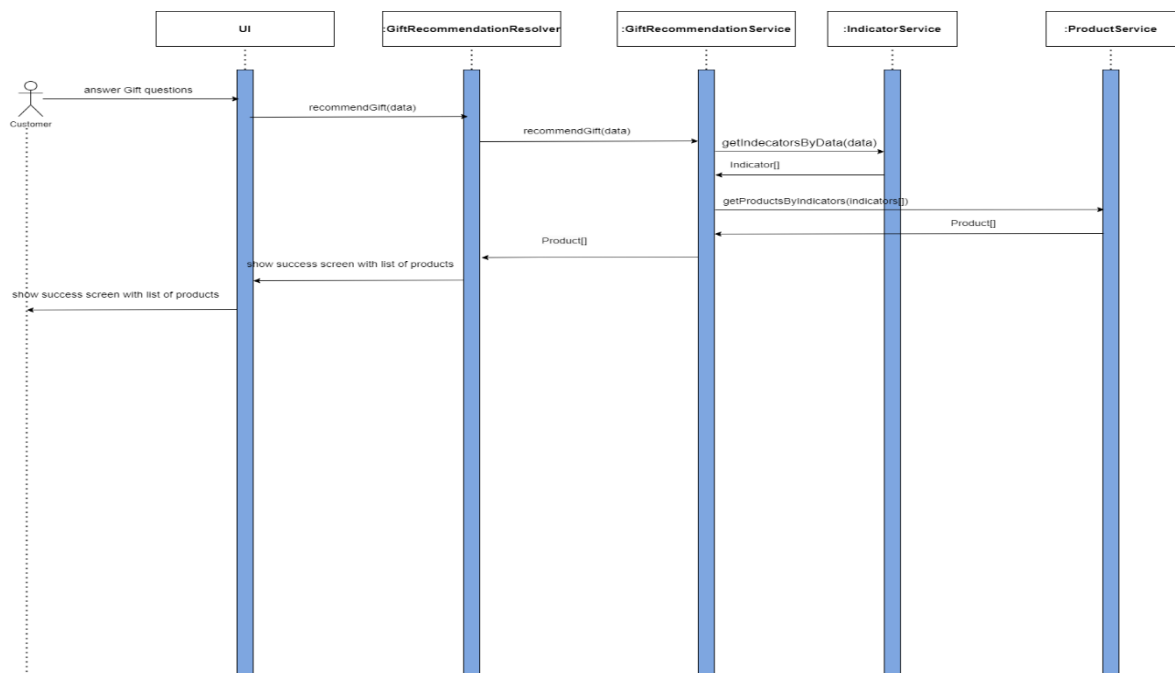


Figure 42: Generate Gift Sequence Diagram

4.3.9 Generate Image

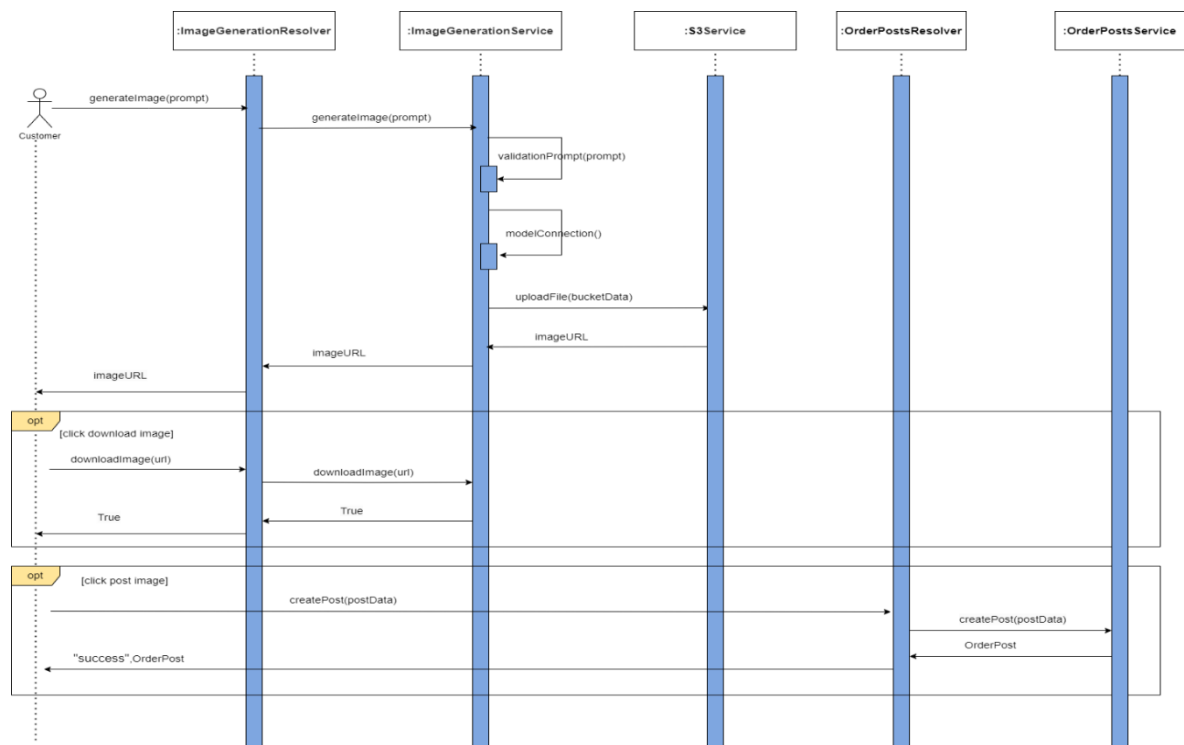


Figure 43: Generate Image Sequence Diagram

4.3.10 Join Bazar

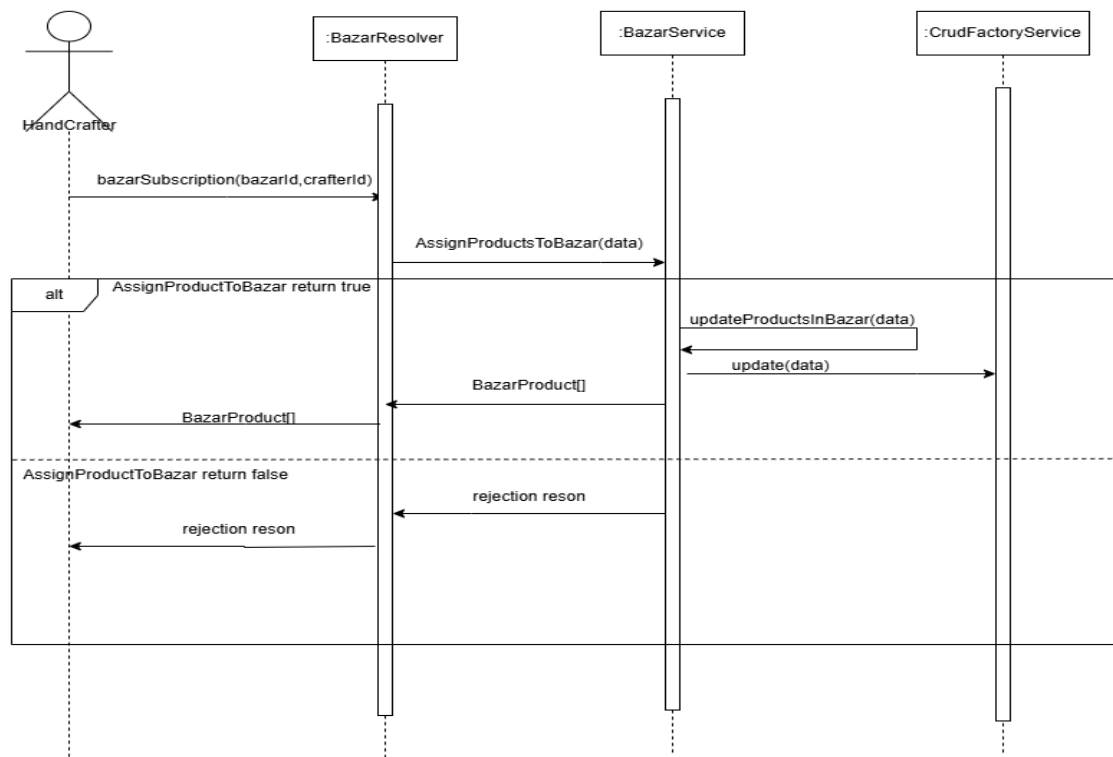


Figure 44: Join Bazar Sequence Diagram

4.3.11 Login

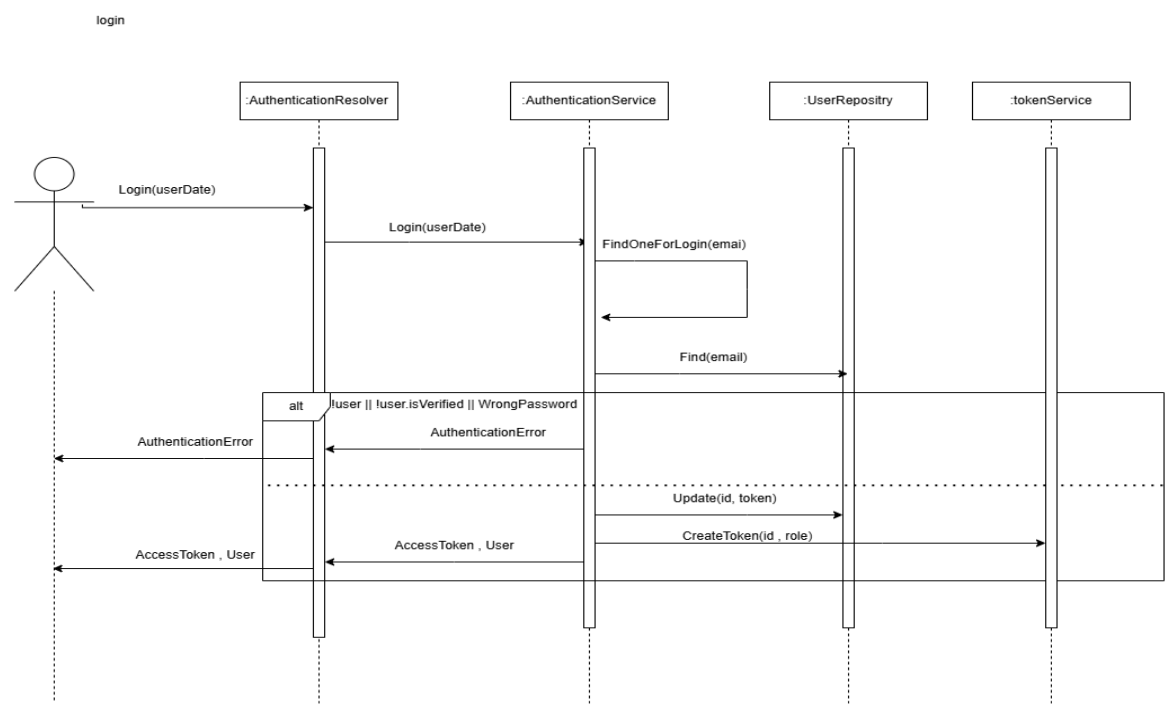


Figure 45: Login Sequence Diagram

4.3.12 Add Reel

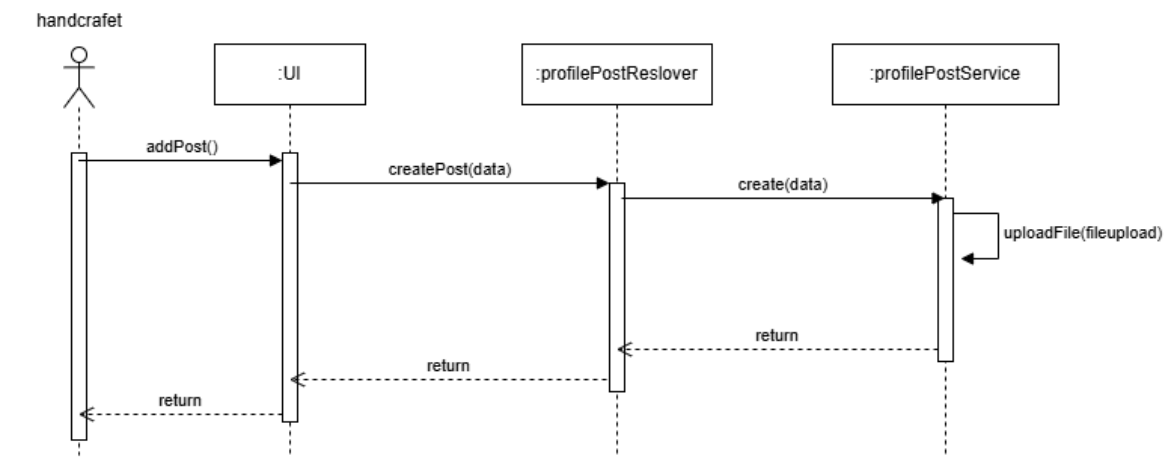


Figure 46: Add Reel Sequence Diagram

4.3.13 Sign Up

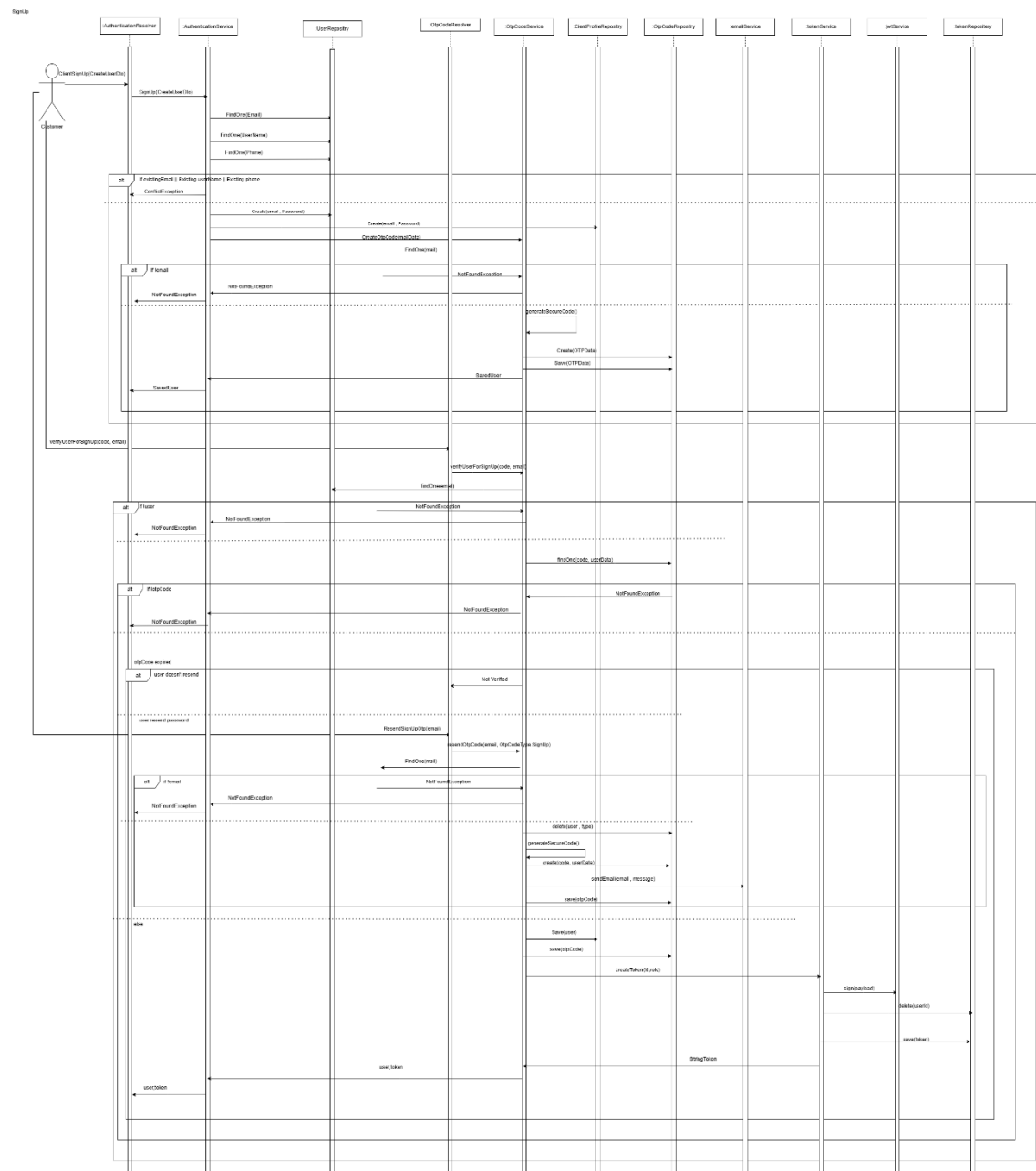


Figure 47: Sign Up Sequence Diagram

4.4 System Architecture

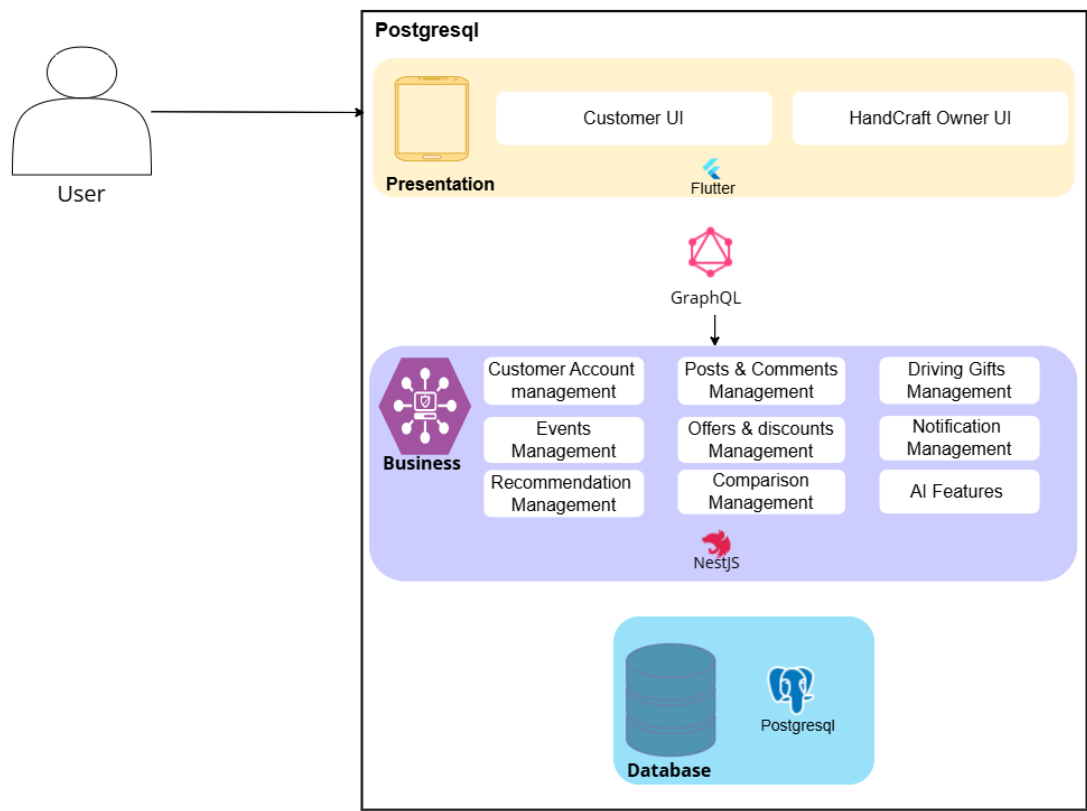
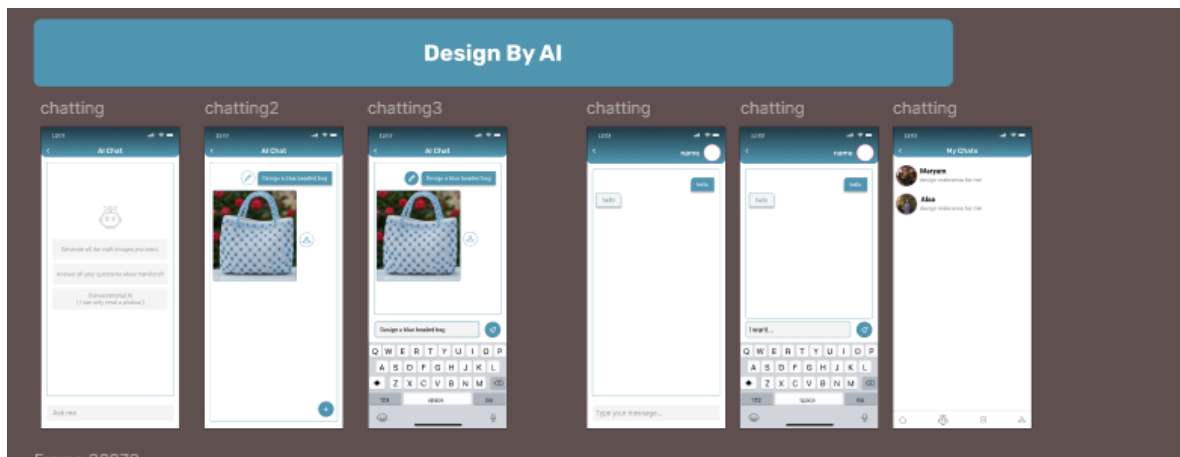
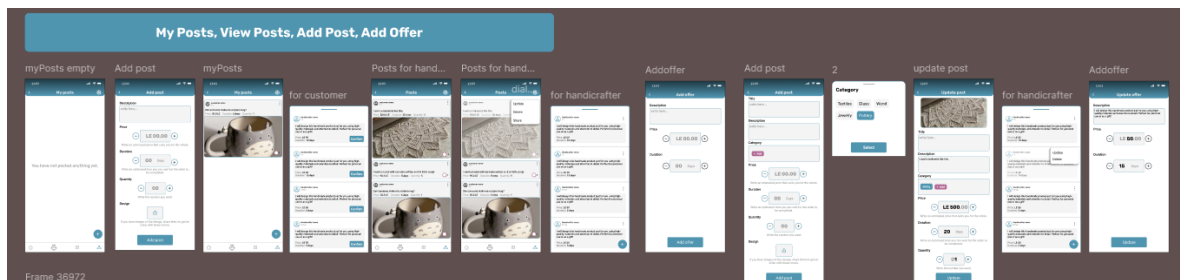
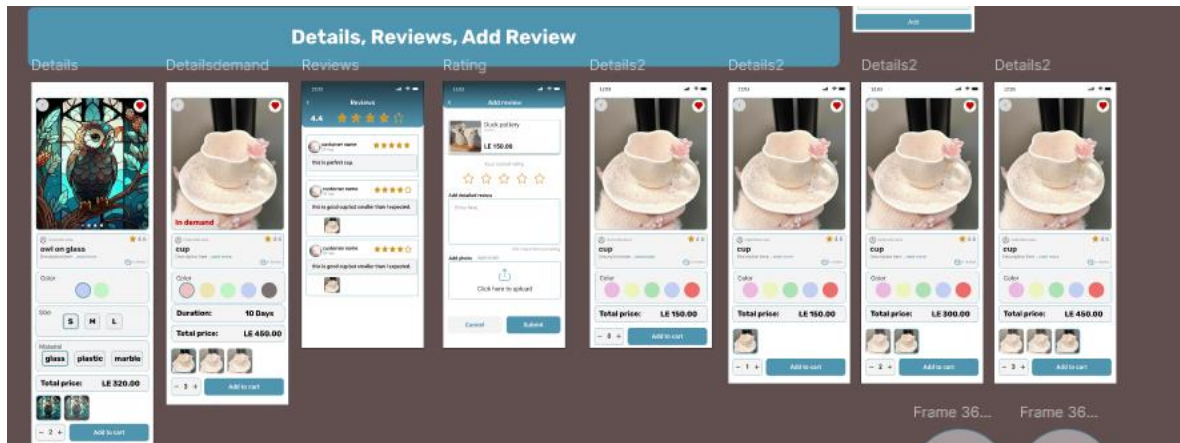
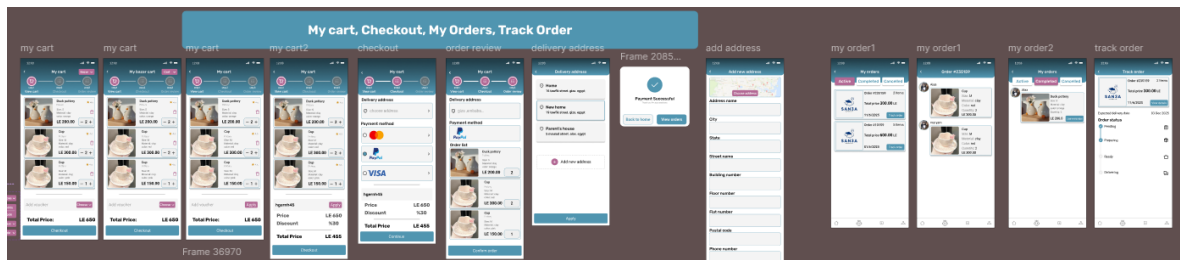
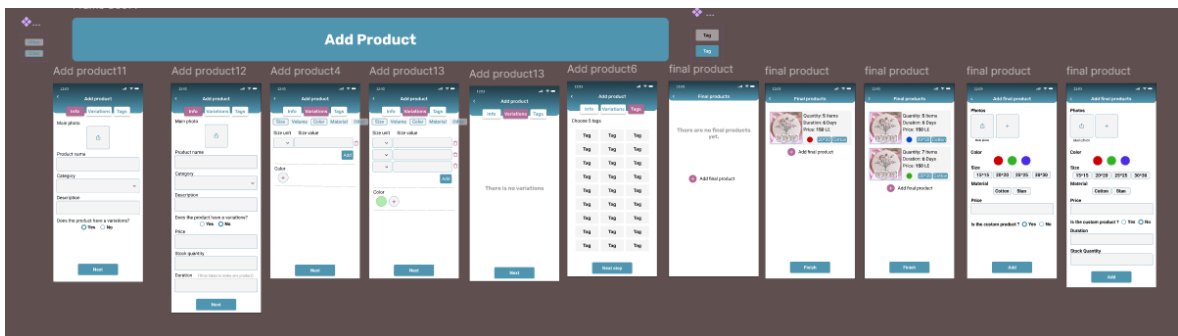
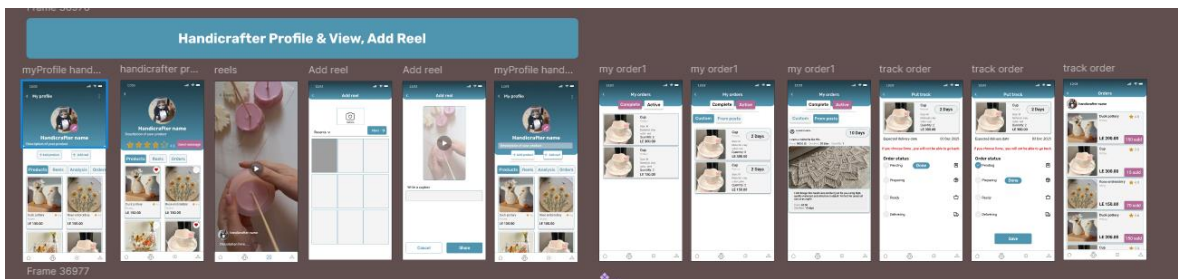
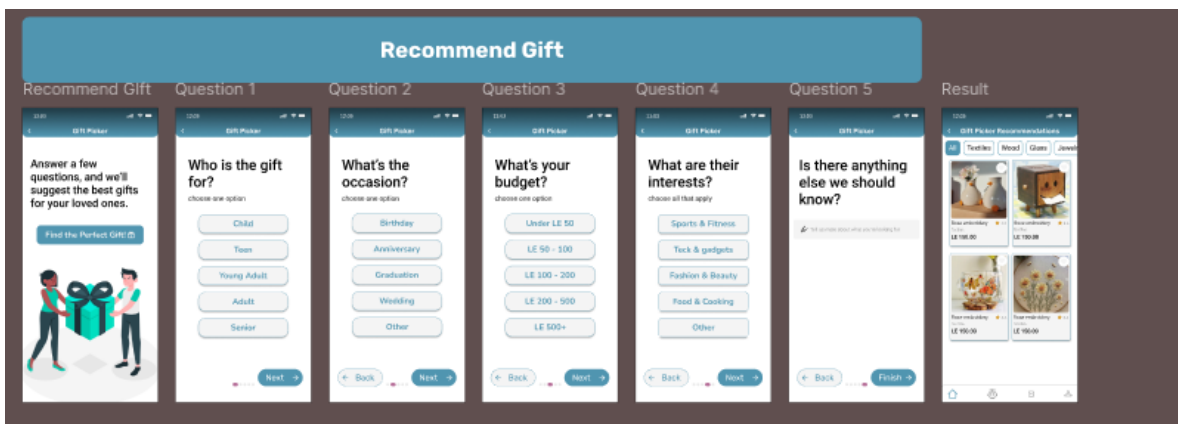
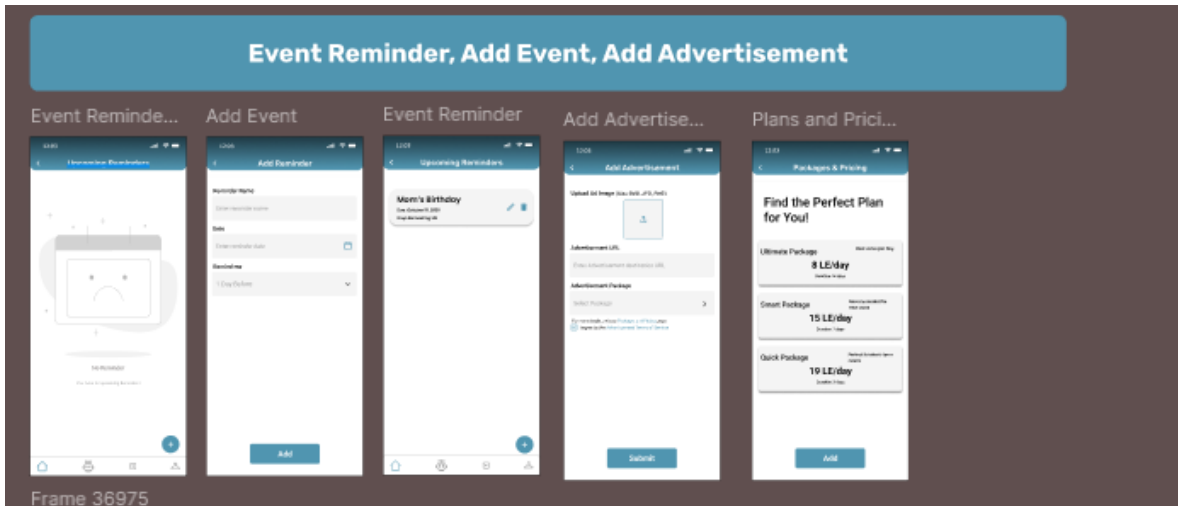


Figure 48: System Architecture

4.5 Figma UI







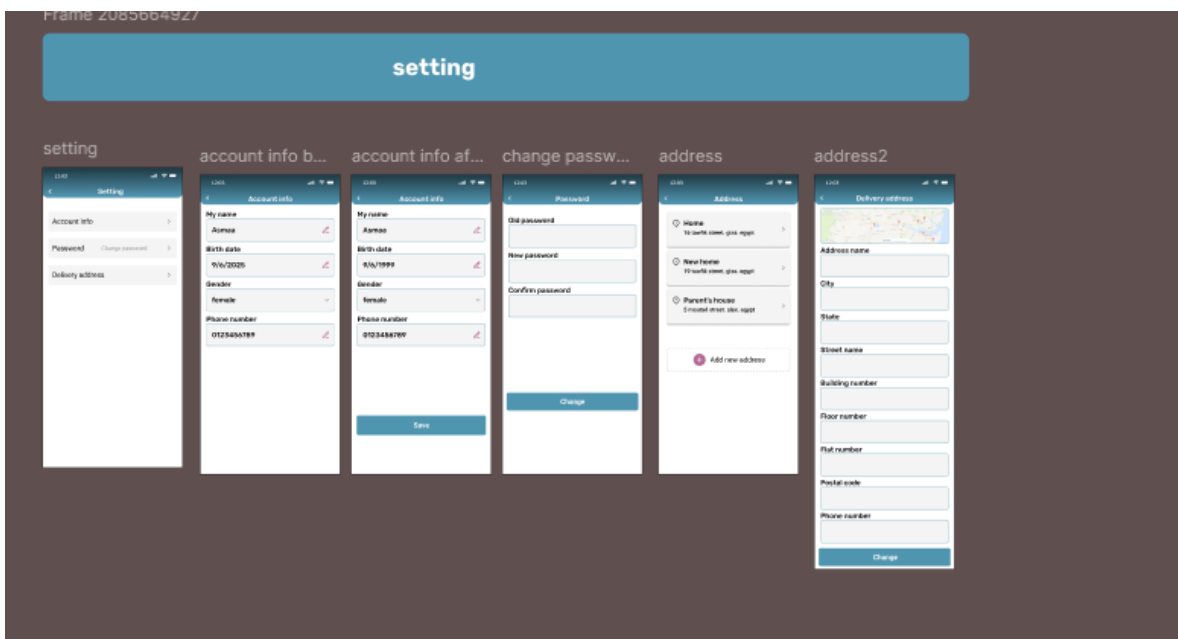
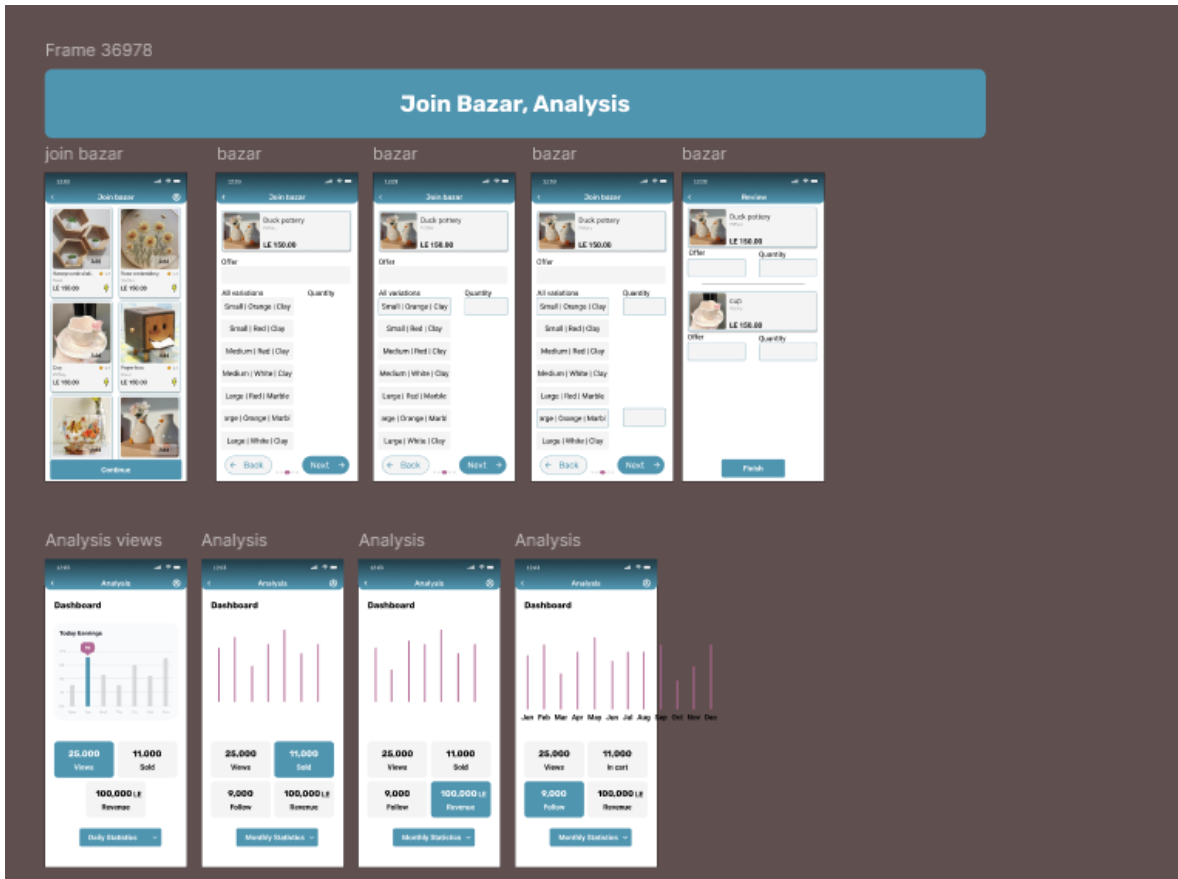


Figure 49: Figma UI

Chapter 5: Implementation and Testing

5.1 Development Environment & Tools

The project was developed using a variety of tools and technologies to support efficient implementation and deployment:

Tool/Platform	Purpose
Flutter	Mobile App Development (Cross-platform)
NestJS	Backend Framework
GraphQL	API Communication between Backend and Frontend
PostgreSQL	Relational Database
DigitalOcean	Cloud Hosting for Backend and Storage
Microsoft Azure	Database Hosting
Docker	Containerizing PostgreSQL Database
Hugging Face	Hosting AI/NLP Models
OpenAI API	Integrating External AI Models
Qdrant	Vector Database for AI-powered Search
Postman	API Testing
GitHub	Version Control and Code Collaboration
Trello	Task & Project Management
Figma	UI/UX Design and Prototyping
Lucidchart / Visual Paradigm	System Design and Architecture Diagrams

5.2 Key Modules Implemented

- **Products Module:**
 - Handles CRUD operations for products, including filtering and sorting.
- **Authentication Module:**
 - Manages user registration, login, and token-based authentication using JWT.
- **Bazar and Advertisement Module:**
 - Allows users and handcraft owners to post and manage product advertisements and bazar listings.
- **Customized Products Module:**
 - Enables users to request and order custom-made products, specifying materials, colors, and size.
- **Notification Module:**
 - Implements push notifications and in-app notifications for orders, promotions, and events.
- **Shared Logs & GiftCodes Module**
 - Tracks important system events and manages promotional gift codes.
- **User Profiles Module:**
 - Manages user information, addresses, and preferences.
- **Orders Module:**
 - Handles placing orders, order tracking, and order history.
- **AI Integrations Module:**
 - Integrates external AI services for:

- Personalized recommendations
- Search enhancements

5.3 Implementation Challenges and Solutions

Challenge:

The system needed to support two distinct types of users: customers and handcraft owners, each with different features, permissions, and access rights. This introduced complexity in user management, authentication, and authorization. It was essential to ensure that:

Customers could only access shopping, browsing, and purchasing features.

Handcraft owners could manage their products, offers, and orders.

Solution:

We implemented role-based access control (RBAC) in the backend using NestJS's Guards and Decorators.

Users were assigned a role (CUSTOMER, HANDCRAFT_OWNER, ADMIN, etc.) during registration or account creation.

Each route in the API was protected using custom Guards that checked the user's role before allowing access.

The authentication process used JWT tokens, where the user's role was embedded as a claim, allowing fast and secure role validation in each request.

Additionally, we structured the database to store user roles

Challenge:

The project integrated several AI-powered features, such as personalized product recommendations and AI-driven search. Running these models efficiently was a challenge due to:

High computational demands of running NLP and recommendation models.

The need to provide low-latency responses, especially for product recommendations and AI-powered search.

Handling large-scale product and user data vectors efficiently.

Solution:

To address this, we used Hugging Face models for generating embeddings and AI tasks like product recommendations and personalized suggestions. These models were pre-loaded and optimized on the backend to minimize loading times.

Qdrant, a vector search engine, was integrated to store product and user embeddings efficiently and perform real-time similarity searches for recommendations. Qdrant allowed us to perform fast nearest-neighbor searches over high-dimensional vectors. Heavy computations were offloaded from the main server and handled asynchronously using background tasks where possible, preventing blocking API requests.

This setup reduced model response time significantly and allowed scalable AI operations without degrading the core app's performance.

5.4 Testing

Unauthorized Access:

Description: A regular Customer tries to access a Hand Crafter-only feature, like adding a product to the bazaar	
Expected: Access denied with a clear error response.	Failure Cause: Missing or incorrect role-based access control in the API.

AI Recommendation Service Timeout

Description: A user opens the Main page, triggering an AI model to generate recommendations.	
Expected: Recommendations are fetched and displayed within a short time.	Failure Cause: AI model takes too long or Qdrant is down, causing a timeout.

Deployment Environment Version Mismatch

Description: Backend is deployed on DigitalOcean with an incorrect Node.js version incompatible with the code.	
Expected: The server runs smoothly with the correct environment versions.	Failure Cause: No Docker containerization; incorrect Node.js version installed.

AI API Key Missing

Description: AI features (e.g., OpenAI or Hugging Face) are called without the API key being properly configured.	
Expected: The request succeeds and AI responses are returned.	Failure Cause: API key missing in environment variables.

Database Connection Failure

Description: The backend tries to access PostgreSQL, but the database server is down or misconfigured.	
Expected: Database queries complete successfully.	Failure Cause: Database not reachable due to network failure or misconfiguration.

GraphQL Syntax Error

Description: A frontend sends an invalid GraphQL query to the backend.	
Expected: The backend parses and executes the query successfully.	Failure Cause: Syntax error in the GraphQL query structure..

Chapter 6: Future Plan

6.1 Future work

- **Wishlist Functionality Improvement**
 - Replace PDF sharing with direct link sharing
 - Social media integration for wishlist sharing
 - Enhanced sharing options across platforms
- **Shipping Company Partnerships**
 - Contract establishment with reliable shipping providers
 - Real-time tracking integration
 - Competitive shipping rate negotiations
 - Multi-region delivery coverage
- **Payment System Expansion**
 - Integration of additional payment gateways
 - Digital wallet support
 - Installment and flexible payment plans
- **Customizable Seller Profiles**
 - Enable artisans to personalize their storefronts
 - Enhanced profile customization options
 - Improved customer attraction features
- **Enhanced Feature Flags**
 - Eco-friendly delivery options
 - Faster delivery for premium users
 - Environmental sustainability features

